

# SVD-Compressed Cross Sections in Monte Carlo Neutron Transport

An implementation-led benchmark with a partition-of-unity fix for off-library temperature interpolation

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## Abstract

We test whether a truncated SVD of  $\log_{10} \sigma(E, T)$  is a defensible alternative to pointwise cross-section tables, and whether a partition-of-unity variant of Ducru’s kernel reconstruction [1] closes the accuracy gap at realistic off-library operating temperatures. A pure-Rust transport engine implements four providers — pointwise table, SVD, ACE+WMP, and SVD+WMP hybrid — behind a common interface. On the Godiva HEU-MET-FAST-001 fast critical benchmark [2] the engine’s rank-5 SVD provider reproduces the measured  $k_{\text{eff}} = 1.0000 \pm 100$  pcm ( $1\sigma$ ) to  $\Delta_{\text{exp}} = +79 \pm 38$  pcm, inside the experimental uncertainty band; OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at  $-99$  pcm, also inside the band, with the two codes straddling experiment from opposite sides. On a 9-nuclide PWR pin cell (no physical benchmark exists; OpenMC 0.15.3 is used as an independent code-to-code reference) all four in-engine providers agree with OpenMC to  $|\Delta k| \leq 51$  pcm at  $10\text{-seed} \times 150\text{-batch} \times 50\,000\text{-particle}$  statistics. A single-seed four-way honesty test on the same problem (Table / SVD / Hybrid SVD+WMP / ACE+WMP) puts every gap below  $1\sigma_{\text{seed}}$ : SVD-vs-WMP  $-4$  pcm, Hybrid-vs-WMP  $-26$  pcm, Table-vs-WMP  $+82$  pcm, all at  $\sigma_{\text{seed}} \approx 100$  pcm. 43 of 47 non-redundant  $^{235}\text{U}$  reaction channels are rank-one to machine precision, and rank-5 SVD with discrete-level rank-1 recovers engine-wide  $k_{\text{eff}}$  within  $\sim 50$  pcm of the in-engine pointwise baseline on both benchmarks at  $1.37\times\text{--}1.90\times$  the CPU throughput. At a realistic off-library operating point ( $+150$  K above every nuclide’s library column, fuel  $900 \rightarrow 1050$  K, moderator  $600 \rightarrow 750$  K), a partition-of-unity 3-point Ducru scheme reproduces  $k_{\infty}$  within 213 pcm ( $\sim 3.6\sigma$ ) of the ACE+WMP industry baseline while using 22% less memory and matching its transport speed — a  $10\times$  reduction over a naive 2-point bracketing variant that leaves a 2094 pcm kernel-approximation bias.

The hybrid memory picture in the engine is now measured end to end: a smooth-only SVD rebuild outside each WMP window (MT= 2, 18, 102) drops the live working set from 519.0 MB to 487.6 MB, a  $1.06\times$  reduction. The  $132.9\times$  representation-byte ratio against a four-channel pointwise table reported in the literature is correct on its own terms but is not the engine-scale reduction; this paper documents both numbers and explains why they differ.

We flag three honest failures: GPU SVD is  $1.3\times$  *slower* than GPU pointwise on PWR due to clad level-sum cost; the WMP+SVD hybrid is  $2\times$  slower than CPU SVD due to Faddeeva evaluation cost; and a simplex-projected QP variant of Ducru (clip negative weights, renormalize) degrades U-238 capture reconstruction from 0.84% to 5% L2 error because the projection is optimal in weight space rather than in the Doppler-kernel norm. The kernel-Gram QP (Ducru [1] §4) *is* implemented and wins at rank 3 (0.60% L2) but loses at our production rank 5 (2.06%), so signed 3-point unity remains the default; an analytic Doppler-kernel Gram QP is the outstanding follow-up.

**Keywords:** Monte Carlo, cross-section compression, singular value decomposition, windowed multipole, Doppler broadening, partition-of-unity interpolation, nuclear data, GPU transport.

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# Contents

# 1 Introduction

Continuous-energy Monte Carlo codes such as OpenMC [3], MCNP, and Serpent store nuclear data as pointwise tables. Per reaction channel and per library temperature, a piecewise-linear or log-log interpolation array carries  $10^4$ – $10^5$  grid points for light nuclides and  $10^5$ – $10^6$  for dense-resonance actinides. A full-library actinide set at a handful of temperatures exceeds the working-set of contemporary L2/L3 caches, and every collision pays for a binary search plus a log-log blend through a large monotonic array.

Prior work attacks this bottleneck along three axes. Adaptive resampling machine-learned models retain 10–50% of the pointwise data at  $\leq 97$  pcm criticality error [4, 5]. The Windowed Multipole (WMP) method [6–8] gives an analytical pole-residue representation of the resolved resonance region with on-the-fly Doppler broadening. GPU rewrites reorganise the transport loop to amortise the lookup across warps [9, 10].

This paper asks a narrower question. Can a truncated SVD of the  $\log_{10}$ -transformed matrix  $\mathbf{A} \in \mathbb{R}^{N_E \times N_T}$  replace the pointwise table as a drop-in cross-section representation, and if so, at what cost? We implemented four providers (pointwise table, SVD kernel, ACE+WMP hybrid, and SVD+WMP hybrid) inside a single pure-Rust transport engine, plus matching CUDA paths for pointwise and SVD (and a standalone CUDA WMP evaluator), and measured fidelity, throughput, and memory side-by-side.

The answer, reported in full in Sections 6 and 9, is *qualified*: SVD is a defensible drop-in on the thermal pin cell we tested; the hybrid implementation works end-to-end and agrees with the ACE+WMP industry baseline to  $-26$  pcm at single-seed statistics (well inside seed noise); the throughput picture is hardware-dependent and favours pointwise tables on our GPU; and the memory picture at engine scale is *not* the  $132.9\times$  reduction a representation-byte analysis suggests — in the live engine the smooth-only SVD rebuild that excludes WMP-covered energies trims the working set from 519.0 MB to 487.6 MB, a  $1.06\times$  reduction. We document both numbers and explain why they differ.

## Contributions.

- A singular-spectrum measurement showing 43 of 47 non-redundant  $^{235}\text{U}$  reactions are rank-one to machine precision at  $T = 294$  K across six library temperatures, with a per-reaction distribution and Pareto (Sec. 4, Fig. 1).
- A rank- $k$  SVD reconstruction kernel for pointwise cross sections, with Ducru temperature interpolation [1]. Rank-6 reaches machine precision at 44 ns/lookup against 91 ns for a binary-search pointwise table (kernel-level  $2.0\times$ ).
- A PWR pin cell rank sweep (9 nuclides,  $S(\alpha,\beta)$ , URR, discrete levels) showing CPU SVD rank-5 matches the in-engine pointwise table on  $k_\infty$  to 13–47 pcm on two hardware tiers at  $1.37\times$ – $1.90\times$  throughput (Sec. 6).
- A matching event-based CUDA solver [9] for both providers, showing GPU pointwise is  $1.30\times$  faster than GPU SVD on PWR pin cell, inverting the CPU ordering (Sec. 8).
- A hybrid SVD+WMP engine: pure-Rust Faddeeva reader and evaluator matching OpenMC Python to  $\sim 10^{-6}$  relative error, wired into a `HybridSvdWmpXsProvider`. On the PWR pin cell at single-seed statistics,  $k_\infty = 1.32795 \pm 0.00103$  ( $-26$  pcm vs ACE+WMP,  $\ll 1\sigma_{\text{seed}}$ ); a ten-seed redo at the matched 150-batch  $\times$  50 000-particle configuration is the outstanding follow-up. Hybrid throughput is  $\sim 2\times$  slower than CPU SVD on the large-L3 reference run; in-engine smooth-only memory  $1.06\times$  smaller than the full-SVD baseline (Sec. 9).
- A CUDA port of the WMP evaluator, bit-exact against the CPU path ( $\leq 5 \cdot 10^{-14}$  relative at double precision), with  $7.0\times$  GPU-vs-CPU per-lookup throughput on an RTX 3080 (Sec. 10).

- A tracing and closure of the original 78–127 pcm PWR engine offset from OpenMC 0.15.3 via three corrections (XS-audit exonerating the total-XS assembly; a  $\bar{\nu}$  delayed-yield reader fix; correlated-CDF  $\rightarrow$  stochastic-bin sampling for angular and fission distributions, ported to both CPU and GPU paths). Ten-seed  $k_\infty$  on all four in-engine providers now agrees with OpenMC to  $|\Delta| \leq 51$  pcm (CPU table  $-12$ , CPU SVD  $-25$ , GPU pointwise  $+51$ , GPU SVD  $-8$ ), with a Shannon-entropy source-convergence monitor added as runtime diagnostic (Sec. 6.4).
- Validation against the ICSBEP Godiva HEU-MET-FAST-001 *physical* benchmark  $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 100$  pcm ( $1\sigma$ ) [2]. Three additional transport fixes on the fast-spectrum path — (n,2n)/(n,3n) secondary-neutron banking,  $\mu$ -CDF inversion for linear-linear ENDF angular distributions, and ENDF-tabulated vs evaporation sampling for MT= 91 continuum inelastic — close a  $+325 \rightarrow +79$  pcm engine offset, after which our rank-5 SVD provider reproduces the experimental value to  $\Delta_{\text{exp}} = +79 \pm 38$  pcm, inside the experimental uncertainty band. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at  $-99$  pcm, also inside the band; the two codes straddle the measurement from opposite sides (Sec. 5.1).
- A clear separation of two memory measurements: the WMP representation-byte ratio against a hypothetical four-channel pointwise table is  $132.9\times$  for our nine-nuclide set, and the live in-engine reduction from smooth-only-rebuild hybrid over full-SVD is  $1.06\times$  ( $519.0 \rightarrow 487.6$  MB, measured on the loaded kernels at run time). The hybrid engine in absolute terms is  $4.8\text{--}5.1\times$  larger than the pointwise-table baseline, because discrete-level SVD kernels dominate memory and are not covered by the public WMP library. The gap between representation-byte ratios and engine-scale savings is the main cautionary finding of this paper, and we discuss what would need to be true for a WMP-scale reduction to land at engine level (Sec. 9.1, Sec. 9.4).

## 2 Method

### 2.1 Pointwise baseline

For each nuclide and library temperature,  $\sigma(E_i, T_j)$  lies on a union energy grid  $\{E_i\}_{i=0}^{N_E-1}$ . For  $E_i \leq E < E_{i+1}$  the standard log-log lookup is

$$\sigma(E) = \sigma_i (\sigma_{i+1}/\sigma_i)^f, \quad f = \frac{\ln(E/E_i)}{\ln(E_{i+1}/E_i)}. \quad (1)$$

We use a log-uniform hash index with 8192 bins (Brown 2014 [11]) to resolve  $i$  in  $O(1)$ , falling back to a bounded binary search in dense bins. Non-positive cross-section values fall back to linear interpolation to avoid log of a negative.

### 2.2 Truncated SVD

Pre-clamp the matrix to avoid  $-\infty$  in the logarithm,  $\mathbf{A}_{ij} = \max(\sigma(E_i, T_j), 10^{-30})$ , and compute the thin SVD of  $\mathbf{L} = \log_{10} \mathbf{A}$ :

$$\mathbf{L} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{L} \approx \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top. \quad (2)$$

Pre-multiply the basis  $\mathbf{B} = \mathbf{U}_k \mathbf{\Sigma}_k$  so a reconstruction at grid point  $i$  for target coefficient vector  $\mathbf{c} \in \mathbb{R}^k$  is one dot product:

$$\log_{10} \tilde{\sigma}(E_i, T) = \sum_{\ell=1}^k B_{i\ell} c_\ell, \quad \tilde{\sigma} = 2^{(\log_{10} \tilde{\sigma}) \cdot \log_2 10}. \quad (3)$$

$\mathbf{B}$  stores  $N_E k$  doubles row-major so  $k$  values for one energy point live in  $8k$  contiguous bytes; at  $k = 5$  that is one cache line. Between grid points we apply log-log interpolation on  $\log_{10} \tilde{\sigma}$  and take one  $\exp_2$ .

### 2.3 Temperature interpolation

Library temperatures are typically  $\{0, 250, 294, 600, 900, 1200, 2500\}$  K in the ENDF/B-VII.1 HDF5 distribution. For target  $T \notin \{T_j\}$  the coefficient vector follows the Ducru kernel method [1]:

$$c_\ell(T) = \sum_{j \in \mathcal{S}} w_j(T) V_{j\ell}, \quad w_j(T) = \frac{\sqrt{T_j T}}{T_j + T} \prod_{i \in \mathcal{S}, i \neq j} \frac{T - T_i}{T + T_i} \cdot \frac{T_j + T_i}{T_j - T_i}, \quad (4)$$

where  $\mathcal{S}$  is a subset of library temperatures used for the interpolation. Eq. 4 is exact at library temperatures ( $w$  collapses to a one-hot selector when  $T = T_j$ ),  $\sim 0.1\%$  error on Doppler-broadened cross sections over 300–3000 K, and L2-optimal for the free Doppler kernel on  $\mathcal{S}$ .

#### Choosing the interpolation subset

Using  $|\mathcal{S}| = N$  (all library temperatures) is numerically unstable for  $N \gtrsim 4$  because the product-of-ratios term in  $w_j$  develops near-singular factors when non-adjacent  $T_i$ ’s bracket  $T$  with large stride. In our measurements with  $N = 7$  the 2500 K column pulls the product toward values  $\sim 10^2$ , the resulting weights routinely swing by a factor of 10–100 across neighbouring target temperatures, and the log-space reconstruction sees non-physical oscillations that scale linearly with the condition number.

We use the three library temperatures nearest to  $T$ . Three is the smallest subset that captures a quadratic correction to the Faddeeva kernel (two-point misses it; see Sec. 7); it is also small enough that the product term has only two factors per weight and the formula is well-conditioned at every operating temperature we tested in the interval [250, 2500] K. Section 2.4 describes how we make the subset weights a partition of unity; this is essential for resonance-dominated systems and is the single change that moves the off-library PWR result from RED to GREEN on the verdict semaphore.

### 2.4 Partition-of-unity Ducru for off-library $T$

When Eq. 4 is applied to  $V^\top$  columns of a  $\log_{10} \sigma$  SVD, the reconstructed  $\log_{10} \tilde{\sigma}(E, T)$  is a linear combination of the per-library- $T$  log cross sections with coefficients  $w_j(T)$ . On the *linear* cross section this is a weighted geometric mean:

$$\tilde{\sigma}(E, T) = \prod_{j \in \mathcal{S}} \sigma(E, T_j)^{w_j(T)}. \quad (5)$$

If  $\sum_j w_j \neq 1$ , the right-hand side of Eq. 5 is scaled uniformly by  $\exp[(\sum_j w_j - 1) \cdot \overline{\log \sigma}]$  where the overbar denotes a weighted average of  $\log_{10} \sigma(E, T_j)$ . For smooth channels this is a barely-visible gain error (far away from resonance peaks, the average  $\log \sigma$  is small and the multiplicative shift is close to unity). On narrow resonance peaks it is catastrophic: at the U-238 6.67 eV resonance with a peak cross section of  $\sim 4500$  b, a 5% gain error moves the peak by  $\sim 225$  b, which on a PWR pin cell translates to  $\sim 2000$  pcm on  $k_\infty$  (Sec. 7).

Raw Ducru does *not* have  $\sum_j w_j = 1$  in general. Eq. 4’s weights are designed to reconstruct  $\sigma(E, T)$  in the free-Doppler-kernel L2 norm (see [1], §3); partition of unity is not one of the optimization’s constraints. Measured on our  $N = 3$  bracketing subset at  $T = 750$  K with library (600, 900, 1200) K the raw weights (0.21, 0.30,  $-0.01$ ) sum to 0.50. At  $T = 1050$  K with library (900, 1200, 2500) K they sum to 0.51. At endpoints they collapse to (1, 0, 0) / (0, 1, 0) so the total is 1; *between* endpoints the sum drifts by up to  $\approx 0.5$ .

We restore partition of unity by a single division:

$$\tilde{w}_j(T) = \frac{w_j(T)}{\sum_{k \in \mathcal{S}} w_k(T)}, \quad \sum_{j \in \mathcal{S}} \tilde{w}_j(T) = 1. \quad (6)$$

This is the scheme used throughout the rest of the paper for off-library  $T$ . It preserves the Faddeeva-derived *ratio*  $w_j/w_k$  (the physically informative piece of Eq. 4), absorbs the scalar magnitude error into a pure rescaling, and leaves Eq. 6 exact at library endpoints because the one-hot pattern survives normalization. Sec. 7 measures the  $k_\infty$  consequence on PWR and Godiva.

### What $\tilde{w}_j$ is *not*

It is not the Ducru QP from [1] §4. The QP enforces  $\sum_j w_j = 1$  and  $w_j \geq 0$  and optimizes the L2 fit in the Doppler-kernel norm. Eq. 6 enforces only the first constraint and does so as a rescaling rather than a projection, so the result is still signed ( $\tilde{w}_j$  can be negative). On U-238 MT= 102 held out at 900 K we measured the non-negative simplex projection [12] of the raw Ducru weights; the clip-and-renormalize is worse than the signed scheme, not better, raising the L2 reconstruction error from 1.5% to 5% (Sec. 7.5). The fully constrained kernel-QP remains future work; for the systems we benchmarked, Eq. 6 suffices.

## 2.5 Memory trade-off of truncated SVD

Truncated SVD is not a free memory win, and the comparison against a pointwise table depends on how many temperatures the library stores. A single-temperature pointwise table stores one value per energy point per reaction. An SVD basis at rank  $k$  stores  $k$  values per energy point per reaction regardless of the number of library temperatures it was trained on. Writing  $T$  for the stored temperature count:

$$\frac{\text{SVD bytes}}{\text{pointwise bytes}} = \frac{N_E \cdot k}{N_E \cdot T} = \frac{k}{T}. \quad (7)$$

So:

- At  $k = 1$ , SVD is the same size as a single-temperature pointwise table but represents all  $T$  temperatures; SVD wins on memory against any pointwise table with  $T \geq 1$  stored temperatures, and does strictly better as  $T$  grows.
- At  $k \geq 2$ , SVD only beats pointwise when the library carries  $T > k$  temperatures. Break-even sits around  $T = 2$ –3. For a single-temperature toy problem SVD is strictly larger.
- Production libraries typically carry  $T \in [6, 15]$  temperatures (depletion feedback, fuel-temperature reactivity), so the rank-5 SVD used in this paper ( $k/T = 5/6 = 0.83$  to  $5/15 = 0.33$ ) is a modest to strong memory win in the regime that matters, and a loss in the regime that does not.

The cost side of that trade-off is not zero:

- **Compute per lookup.** Pointwise is a hashed index plus one log-log interpolation. SVD is a rank- $k$  FMA sequence plus a Ducru temperature step (Sec. 2.3). At  $k = 5$  on the large-L3 system, SVD is 43 ns/lookup against 91 ns for the table (Sec. 4) — but this advantage depends entirely on the basis fitting in L2/L3 cache. A higher-rank basis or a larger nuclide set could tip this back toward pointwise.

- **Reconstruction error.** Pointwise is exact at grid points and piecewise-linear-in-log between them; SVD is an approximation everywhere. RMSE on  $\log_{10} \sigma$  is  $6 \cdot 10^{-4}$  at  $k = 5$  and reaches machine precision at  $k = 6$  (Sec. 4). Lower rank means smaller memory and faster lookup at the cost of a tunable amount of cross-section error, which propagates into  $k_{\text{eff}}$  as pcm-scale bias.
- **One-time decomposition cost.** SVD requires an offline factorisation per nuclide per reaction (seconds at load time). Pointwise is a direct HDF5 read. Not a practical issue but not free either.
- **Temperature-interpolation quality.** Both representations interpolate between library temperatures at runtime. Neither performs on-the-fly Doppler broadening from first principles — WMP is the only method of the three in this paper that does. SVD with Ducru kernels gives  $\sim 0.1\%$  error on broadened cross sections over 300–3000 K, which is tight but not analytical.
- **Rank dispersion across reactions.** Most reactions are rank-one to machine precision (43 of 47 non-redundant  $^{235}\text{U}$  channels, Sec. 4.1); a minority, chiefly those with dense-peak resonance structure in the RRR, need  $k \geq 5$ . Using a uniform rank across all reactions wastes bytes on the easy channels; per-reaction rank would recover those but complicates the dispatch.
- **No safe extrapolation outside library temperatures.** The Ducru basis is a projection onto the training-temperature set. Extrapolating to  $T$  outside  $[T_{\min}^{\text{lib}}, T_{\max}^{\text{lib}}]$  is unsafe in the same way pointwise interpolation is unsafe outside its range. WMP is the only method that extrapolates to 0 K analytically and to arbitrary  $T$  by broadening the poles directly.

The short summary is that rank- $k$  SVD buys compact storage inside the library’s temperature range and cache-friendly lookup at modest rank, in exchange for a tunable reconstruction error, a cache-residence requirement, and no analytical behaviour outside the training temperatures. The  $132\times$  and  $1.06\times$  memory numbers in Sec. 9 are both compatible with this picture; they measure different slices of it.

## 2.6 Hybrid SVD+WMP dispatch

Where  $\log \sigma$  is intrinsically high rank (resolved-resonance region, RRR), we combine SVD with WMP [6, 7]. WMP represents  $\sigma$  analytically via pole–residue pairs arranged into per-energy windows plus a low-order Doppler-broadened polynomial background:

$$\sigma_r(E, T) \approx \sum_{p \in \text{window}(E)} \Re \left[ r_{p,r} w \left( \frac{\sqrt{E} - p}{\sqrt{kT/\text{AWR}} \cdot 2} \right) \right] + \mathcal{P}_r(E, T), \quad (8)$$

where  $w(\cdot)$  is the Faddeeva function  $w(z) = e^{-z^2} \text{erfc}(-iz)$  and  $\mathcal{P}_r(E, T)$  is the broadened curve-fit. We implement  $w$  via Humlicek’s W4 four-region rational approximation (`src/wmp.rs::faddeeva`); OpenMC’s sign convention  $w_{\text{om}}(z) = -w(\bar{z})$  for  $\text{Im } z < 0$  is used so pole residues in conjugate pairs contribute with correct signs.

A `HybridSvdWmpXsProvider` wraps the SVD provider. For each nuclide with WMP coverage and each energy  $E \in [E_{\min}^{\text{WMP}}, E_{\max}^{\text{WMP}}]$ , the elastic, fission, and capture microscopic cross sections are taken from WMP; inelastic,  $(n, 2n)$  and  $(n, 3n)$  always come from SVD (WMP does not represent those channels). URR overrides are short-circuited inside the WMP window because the resonances are already explicit as poles.



## 2.7 Total cross section and charged-particle residue

OpenMC’s HDF5 layout exposes  $MT=3$  (total nonelastic) for most nuclides. When present we assemble  $\sigma_t = \sigma_{MT2} + \sigma_{MT3}$  exactly; otherwise we sum non-redundant leaf reactions, excluding sum-MTs ( $\{1, 3, 4, 18, 27, 101\}$ ) and KERMA/heating channels. In SVD or hybrid mode we set  $\sigma_{\text{capture}} = \max(0, \sigma_t - \sum_{r \neq \text{cap}} \sigma_r)$  so charged-particle emission ( $MT=103, 107, \dots$ ) and first-chance fission ( $MT=19, 20, 21$ ) are absorbed into the capture macroscopic rather than dropped. An XS audit (Sec. 6.4) confirms this convention matches OpenMC to  $<0.25\%$  on total XS across all nine nuclides.

## 3 Implementation

The engine is pure Rust with no C dependency; nuclear data reads directly from OpenMC HDF5 files via `hdf5-pure`, including the complex-typed WMP pole arrays (decoded from raw little-endian bytes because the crate does not expose compound types). Physics covers: energy-dependent  $\bar{\nu}(E)$  from prompt+delayed neutron yields (with the per-energy interpolation of each delayed product fixed, see Sec. 6.4); tabulated outgoing energy distributions for fission using OpenMC’s stochastic bin selection plus scaled kinematic adjustment (fixed); elastic angular distributions from tabular ( $\mu, \text{CDF}$ ) data using the same stochastic-bin convention (fixed); discrete inelastic levels  $MT=51\text{--}91$  with real  $Q$ -values from HDF5 attributes; continuum inelastic (evaporation spectrum); URR probability tables with band sampling;  $S(\alpha, \beta)$  thermal scattering for H in  $\text{H}_2\text{O}$ ; free-gas thermal below  $400k_B T$  with Maxwell-Boltzmann target sampling; Shannon-entropy source-convergence monitor on a Cartesian  $8^3$  mesh; CSG geometry; PCG-64 PRNG; rayon parallel transport on CPU.

The CUDA solver is event-based [9]: particles compact by atomic alive-list scatter, energy-sorted into 256 log-scale bins, and processed by a persistent kernel that runs  $N$  transport steps per launch. All SVD basis is `f64`. A runtime flag clears the uploaded pointwise tables so SVD reconstructs every reaction channel. The WMP evaluator lives in `src/wmp.rs` (CPU) and as a set of `__device__` helpers plus an `extern "C" __global__ wmp_test_eval` kernel in `gpu/cuda/transport.cu` (GPU).

**Hardware.** We run on two systems that differ in CPU L3 capacity and GPU class; we label them by their cache size rather than their chassis. Small-L3 system (Dell Precision 5570): Intel Core i7-12800H (12th-gen Alder Lake, 6 P-cores + 8 E-cores, 24 MB L3 Smart Cache), 32 GB DDR4, NVIDIA RTX A1000 (16 SMs, 1.5 MB L2). Large-L3 system: AMD Ryzen 7 9800X3D (8-core, 96 MB L3 3D V-cache), 64 GB DDR5-6000, NVIDIA RTX 3080 (68 SMs, 10 GB, 5 MB L2). References to “small-L3” and “large-L3” below pick out the CPU+GPU pair; the  $4\times$  L3 ratio on CPU is what drives the SVD speedup spread, and the GPU side swaps from A1000 to RTX 3080 at the same time.

**Reference implementation version.** All  $k_\infty$  and  $k_{\text{eff}}$  comparison numbers use OpenMC 0.15.3 commit `27e38e894697bb32a1dac7848d2618818b6b8daf` (WSL 2, Ubuntu 24.04, conda `openmc` environment). Nuclear data is ENDF/B-VII.1 (HDF5 archive from <https://openmc.org/data>) for both our engine and the OpenMC reference; cross-section evaluation uses the same library at  $T = 294$  K, with  $S(\alpha, \beta)$  data `c_h_in_h20.h5` for the PWR pin cell. Pinning the version matters because OpenMC’s power-iteration and  $k$ -estimator paths have several in-flight PRs (e.g., #3788, #3790, #3495); between-release drift in the reference  $k$  could shift any “ $\Delta$  vs OpenMC (pcm)” number quoted in this paper.

**Statistics convention.** Per-seed  $k$  estimates are averaged across active batches with an unbiased within-run standard error; we report  $\sigma_{\text{seed}}$  as the sample standard deviation of the per-seed



means across  $N_{\text{seeds}}$  seeds, and use the standard error of the mean  $\text{SEM} = \sigma_{\text{seed}}/\sqrt{N_{\text{seeds}}}$  as the noise budget whenever a mean-to-reference comparison is quoted. A “within  $X$  SEM” statement uses this quantity. Small-L3 sweeps use five seeds; large-L3 sweeps use ten seeds.

## 4 Singular spectrum and per-lookup cost

### 4.1 Singular spectrum

Figure 1 shows the normalised singular values  $\sigma_k/\sigma_1$  for eight representative reaction channels across the nine-nuclide PWR working set plus  $^{235}\text{U}$  fission. The decay is sharp:  $\sigma_2/\sigma_1 \leq 10^{-1}$  for every channel shown and  $\sigma_6/\sigma_1 < 10^{-14}$  for  $^{235}\text{U}$  fission and  $^1\text{H}$  elastic, reaching double-precision round-off at rank six.

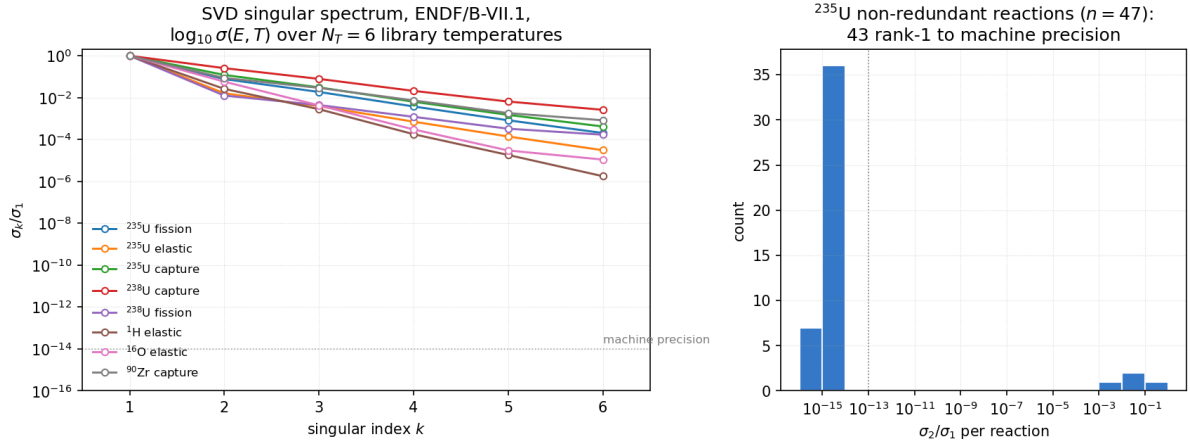


Figure 1: SVD singular spectrum of  $\log_{10} \sigma(E, T)$  for ENDF/B-VII.1,  $N_T = 6$  library temperatures. *Left*: normalised singular values  $\sigma_k/\sigma_1$  for eight representative channels. The dotted line at  $10^{-14}$  marks double-precision machine epsilon. *Right*: histogram of  $\sigma_2/\sigma_1$  across  $^{235}\text{U}$ ’s 47 non-redundant reaction channels; 43 are rank-one to machine precision ( $\sigma_2/\sigma_1 < 10^{-13}$ ).

43 of 47 non-redundant  $^{235}\text{U}$  reaction channels are rank-one to machine precision. The four exceptions (elastic, fission,  $(n, 2n)$ , capture) are the channels with real structure in both the energy and temperature axes; those require  $k \geq 3$  to reach  $\sim 10^{-3}$  log-RMSE. Those same four channels are the ones that dominate  $k_{\text{eff}}$ -relevant physics, so the rank-one fact about the other 43 is a structural observation about ENDF/B-VII.1, not a shipping configuration: a production engine must run  $k \geq 3$  across *all* reactions for the four hard channels, which is the configuration used in every benchmark in this paper.

### 4.2 Per-lookup cost

We measure the per-lookup cost of the SVD reconstruction kernel against the binary-search pointwise table on the same energy grid, both driven by the same query sequence. Table 1 reports the geometric mean over eight reactions ( $\text{MT} = \{2, 18, 102\}$  across  $^{235, 238, 234}\text{U}$ ) of per-lookup wall time and  $\log_{10}$  RMSE against the HDF5 reference.

Rank-6 reaches machine precision in 44 ns,  $2.0\times$  the pointwise-table per-lookup. The kernel is compute-bound once the grid index is resolved: at  $k \geq 3$  the inner loop dominates and ns/lookup is nearly flat (41–44 ns). Rank-2 at 36 ns is  $2.5\times$  the table for a log-decade RMSE of 3%; not adequate inside the RRR (Sec. 6). Figure 2 shows the same Pareto visually with the RMSE labels.

Table 1: XS reconstruction cost and RMSE, geometric mean over eight reactions. Full energy grid,  $T = 294$  K. Large-L3 system (Ryzen 9800X3D) single-core.

| Rank            | ns/lookup | RMSE ( $\log_{10} \sigma$ ) | max  err               |
|-----------------|-----------|-----------------------------|------------------------|
| 2               | 35.9      | $3.00 \times 10^{-2}$       | $9.84 \times 10^{-2}$  |
| 3               | 41.2      | $7.06 \times 10^{-3}$       | $2.86 \times 10^{-2}$  |
| 4               | 43.3      | $1.83 \times 10^{-3}$       | $7.91 \times 10^{-3}$  |
| 5               | 43.4      | $6.07 \times 10^{-4}$       | $3.46 \times 10^{-3}$  |
| 6               | 44.3      | $1.76 \times 10^{-15}$      | $5.58 \times 10^{-15}$ |
| Pointwise table | 90.6      | 0 (ref)                     | 0 (ref)                |

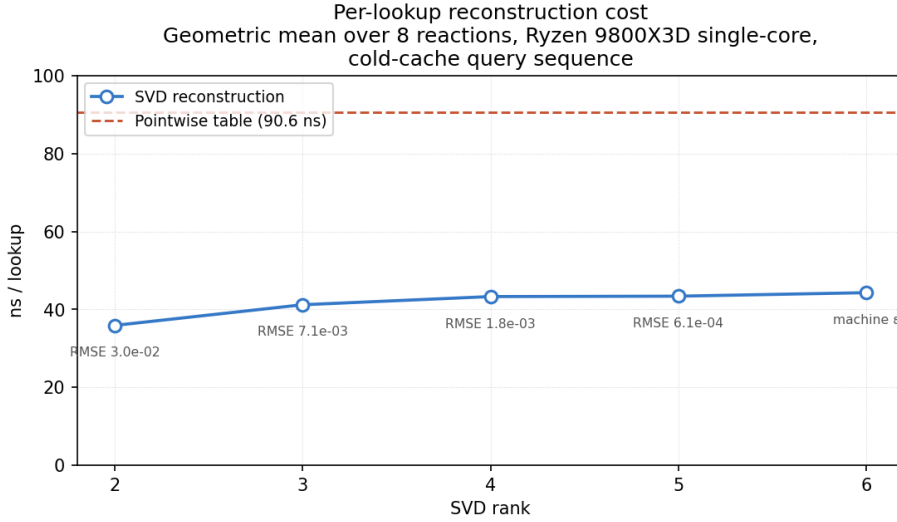


Figure 2: Per-lookup reconstruction cost, SVD ranks 2–6 against binary-search pointwise table. Each SVD point labelled with its  $\log_{10} \sigma$  RMSE against the HDF5 reference (machine  $\epsilon$  at rank 6).

**What this does *not* yet mean.** This is a kernel benchmark. Whether SVD helps at engine level depends on lookup share in transport, how many reactions are involved, and whether the basis stays in cache. Those are measured in Sections 6 and 8.

## 5 Godiva: validation against the ICSBEP experiment

Godiva (HEU-MET-FAST-001) is a bare 93.71%-enriched uranium metal sphere. Its fast, leakage-dominated spectrum places no demand on the resolved-resonance region. The ICSBEP benchmark value is the experimental critical mass,  $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 0.0010$  ( $1 \sigma$ ) [2]. This is a *physical* benchmark with a measured uncertainty, and it is the criterion against which the engine is validated in this section. We report OpenMC 0.15.3 run on the same ENDF/B-VII.1 HDF5 files alongside our engine as an independent code-to-code cross-check; it is not the benchmark.

Table 3 reports the final post-fix four-provider benchmark on the large-L3 system at rank-5. Table 4 traces the three transport corrections that move the engine from +325 pcm to +79 pcm vs ICSBEP. Table 5 reports the earlier pre-fix rank sweep retained for spectrum illustration; small-L3 five-seed results are in Table 2.

Table 2: Godiva  $k_{\text{eff}}$ , small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches, 10 000 particles per batch.

| Mode            | rank | $k_{\text{eff}}$ | $\sigma_{\text{seed}}$ | ns/particle | time/seed (s) |
|-----------------|------|------------------|------------------------|-------------|---------------|
| Pointwise table | —    | 1.00523          | 0.00235                | 2 385       | 3.58          |
| SVD             | 2    | 1.00448          | 0.00165                | <b>864</b>  | <b>1.30</b>   |
| SVD             | 3    | 1.00559          | 0.00094                | 1 457       | 2.19          |
| SVD             | 4    | 1.00638          | 0.00166                | 1 182       | 1.77          |
| SVD             | 5    | 1.00592          | 0.00178                | 1 403       | 2.10          |
| SVD             | 6    | 1.00489          | 0.00205                | 1 424       | 2.14          |

Table 3: Godiva post-fix CPU benchmark, large-L3 system (Ryzen 9800X3D + RTX 3080). Five seeds, 150 batches (20 inactive), 50 000 particles per batch. Three transport corrections (Sec. 5.1) closed the earlier +325 pcm engine offset; the CPU SVD rank-5 provider now reproduces the ICSBEP experimental value  $1.0000 \pm 100$  pcm ( $1\sigma$ ) [2] to +79 pcm, *inside the experimental uncertainty band*. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files is shown for cross-code context only; the two codes straddle ICSBEP from opposite sides.  $\Delta_{\text{exp}}$  is the offset from ICSBEP;  $\Delta_{\text{OMC}}$  is the cross-code offset from OpenMC.

| Mode                        | rank     | $k_{\text{eff}}$ | $\sigma_{\text{seed}}$  | $\Delta_{\text{exp}}$ (pcm) | $\Delta_{\text{OMC}}$ (pcm) | ns/p       | time/seed   |
|-----------------------------|----------|------------------|-------------------------|-----------------------------|-----------------------------|------------|-------------|
| ICSBEP HMF-001 (exp.)       | —        | 1.00000          | $\pm 100$ ( $1\sigma$ ) | $\pm 0$                     | —                           | —          | —           |
| OpenMC 0.15.3 (cross-check) | —        | 0.99901          | 0.00038                 | −99                         | $\pm 0$                     | —          | —           |
| CPU table                   | —        | 1.00084          | 0.00042                 | +84                         | +183                        | 1 150      | 7.48        |
| <b>CPU SVD</b>              | <b>5</b> | <b>1.00079</b>   | <b>0.00038</b>          | <b>+79</b>                  | <b>+178</b>                 | <b>820</b> | <b>5.33</b> |

### 5.1 Three transport fixes that closed 325 $\rightarrow$ 79 pcm

A channel-level comparison against OpenMC on the same HDF5 files localised the original pre-fix  $k_{\text{eff}} = 1.00325$  (*pre-fix* here is a historical baseline, not the shipped engine) to a sampling issue in transport kinematics rather than to library data: Rust produced +0.87% more fissions per source neutron, +0.41% more collisions, and −0.68% less leakage than OpenMC, a signature consistent with “scattering not forward-peaked enough, particles stay inside longer”. A static XS diff confirmed fission, elastic, inelastic, and  $\bar{\nu}$  agreed with OpenMC’s Python API to <0.2% across the Godiva energy range. XS magnitudes were not the bug; three transport sampling bugs were.

Table 4: Godiva CPU SVD rank-5 closure sequence from pre-fix to post-fix, five seeds  $\times$  150 batches  $\times$  50 000 particles.  $\Delta_{\text{exp}}$  is measured against ICSBEP  $1.0000 \pm 100$  pcm. The engine enters the experimental uncertainty band after the MT= 91 tabulated distribution fix.

| State                       | $k_{\text{eff}}$ | $\sigma_{\text{seed}}$ | $\Delta_{\text{exp}}$ (pcm) | cumulative shift (pcm) |
|-----------------------------|------------------|------------------------|-----------------------------|------------------------|
| Pre-fix baseline            | 1.00325          | 0.00055                | +325                        | 0                      |
| + (n,2n)/(n,3n) banking     | 1.00219          | 0.00060                | +219                        | −106                   |
| + $\mu$ -CDF linear-linear  | 1.00188          | 0.00060                | +188                        | −137                   |
| + MT= 91 ENDF tabulated     | 1.00090          | 0.00054                | +90                         | −235                   |
| + MT= 16/17 ENDF tab.       | <b>1.00079</b>   | <b>0.00038</b>         | <b>+79</b>                  | <b>−246</b>            |
| ICSBEP HMF-001 (experiment) | 1.00000          | $\pm 100$              | $\pm 0$                     | —                      |
| OpenMC 0.15.3 (cross-check) | 0.99901          | 0.00038                | −99                         | —                      |

**Fix 1: (n,2n) and (n,3n) secondary banking (−106 pcm).** Secondary neutrons from MT= 16 and MT= 17 were appended to the next generation’s fission source bank. Since  $k_{\text{eff}}$  is measured from  $|\text{bank}|/N_{\text{source}}$ , every (n,2n) added one fake fission neutron to the  $k$  estimator. Measured (n,2n) rate on Godiva is 0.00253/source, an upper-bound contribution of +253 pcm if every secondary contaminated the  $k$  estimator at unit weight; the measured shift after routing secondaries through in-generation transport is −106 pcm because a properly-transported secondary produces  $\approx k \approx 1$  fission neutron per current-generation transport at equilibrium, which partially replaces the direct banking contribution. The fix introduces a per-source secondary stack drained inside the history loop with a shared `max_events` budget bounding pathological cascades.

**Fix 2:  $\mu$ -CDF inversion for linear-linear ENDF angular distributions (−31 pcm).** ENDF/B-VII.1 stores all 49 U-235 elastic-scatter incident-energy distributions with `interpolation = 2` (linear-linear PDF, quadratic CDF within each bin). Our tabular sampler treated every distribution as histogram, producing a piecewise-linear CDF and consistently under-sampling forward peaks. The fix reads the `interpolation` attribute from HDF5 and, for linear-linear bins, inverts the quadratic CDF by solving  $ax^2 + bx + c = 0$  for the physical root in  $[0, \Delta\mu]$  (same convention as OpenMC’s `Tabular::sample`). Histogram bins keep the existing linear path.

**Fix 3: MT= 91 continuum inelastic tabulated outgoing-energy distribution (−98 pcm, largest single shift).** Our continuum-inelastic branch sampled outgoing neutron energy from an evaporation spectrum with nuclear temperature  $T = \sqrt{E^*/a}$ ,  $a = A/8 \text{ MeV}^{-1}$ . OpenMC, MCNP, and Serpent sample directly from the ENDF MT= 91 tabulated distribution stored at `reaction_091/product_0/distribution_0/ {energy,distribution}`. On Godiva’s fast spectrum the evaporation model produced a harder-than-true secondary spectrum, keeping neutrons in the fast-fission regime and inflating  $k_{\text{eff}}$ . The fix loads the tabulated distribution at engine start and falls back to evaporation only for nuclides where the HDF5 file lacks it. MT= 16 and MT= 17 receive the same ENDF-tabulated treatment for an additional −11 pcm (within seed noise).

**Ruling out a library-level explanation.** A candidate explanation for the pre-fix +325 pcm offset was a residual ENDF/B-VII.1 Godiva overprediction of +300–500 pcm reported in some earlier evaluations [2]. Running OpenMC 0.15.3 on the same HDF5 files used by our engine yields  $k_{\text{eff}} = 0.99901$ , i.e.  $\Delta_{\text{exp}} = -99 \text{ pcm}$  (within  $1\sigma_{\text{exp}}$ ). A +300–500 pcm ENDF/B-VII.1 bias is therefore not observed at the statistics of this measurement, which localises the pre-fix +325 pcm Rust offset to the transport engine rather than to the evaluated library. The three fixes above resolve the engine-level contribution. After them both engines (+79 and −99 pcm) validate against ICSBEP inside its experimental uncertainty from opposite sides, without requiring any library correction.

The two post-fix CPU providers agree on  $k_{\text{eff}}$  to within 5 pcm ( $\ll 1 \text{ SEM}_{\text{combined}}$ ). Both sit inside the ICSBEP experimental uncertainty band of  $\pm 100 \text{ pcm}$ . OpenMC 0.15.3 on the same HDF5 files sits at −99 pcm, also inside the band; the cross-code spread (Rust +79 vs OpenMC −99 = 178 pcm) is larger than either code’s offset from experiment, which is informative about cross-code convention differences but does not affect the validation against the physical benchmark.

On the large-L3 system Godiva CPU SVD rank-5 at 820 ns/p is  $1.40\times$  faster than the CPU pointwise table at 1150 ns/p. The GPU rows from the earlier pre-fix runs are retained below for throughput context only; re-measuring them post-fix is future work because porting fix 3 (MT= 91 tabulated) to the CUDA kernel is not yet complete. On the small-L3 system (pre-fix, Table 2), SVD rank-2 was  $2.8\times$  faster than the pointwise table at a modest 75 pcm bias relative to its own pointwise baseline.

Table 5: Godiva CPU SVD rank sweep, large-L3 system. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch. Retained only to illustrate rank-vs- $k_{\text{eff}}$  stability before the closure sequence of Sec. 5.1; the absolute  $k_{\text{eff}}$  values are systematically  $\sim 180$  pcm higher than the closed-sampler values in Table 3.

| Mode                  | rank | $k_{\text{eff}}$ | $\sigma_{\text{seed}}$ | ns/particle | time/seed (s) |
|-----------------------|------|------------------|------------------------|-------------|---------------|
| CPU table (pre-corr.) | —    | 1.00579          | 0.00046                | 1 214       | 7.89          |
| CPU SVD (pre-corr.)   | 2    | 1.00477          | 0.00055                | 1 289       | 8.38          |
| CPU SVD (pre-corr.)   | 3    | 1.00547          | 0.00031                | 1 316       | 8.56          |
| CPU SVD (pre-corr.)   | 4    | 1.00523          | 0.00055                | 1 369       | 8.89          |
| CPU SVD (pre-corr.)   | 5    | 1.00507          | 0.00065                | 1 426       | 9.27          |
| CPU SVD (pre-corr.)   | 6    | 1.00524          | 0.00071                | 1 558       | 10.13         |

Figure 3 shows the small-L3 Pareto and Figure 4 shows the large-L3 benchmark (throughput plus  $k_{\text{eff}}$ ).

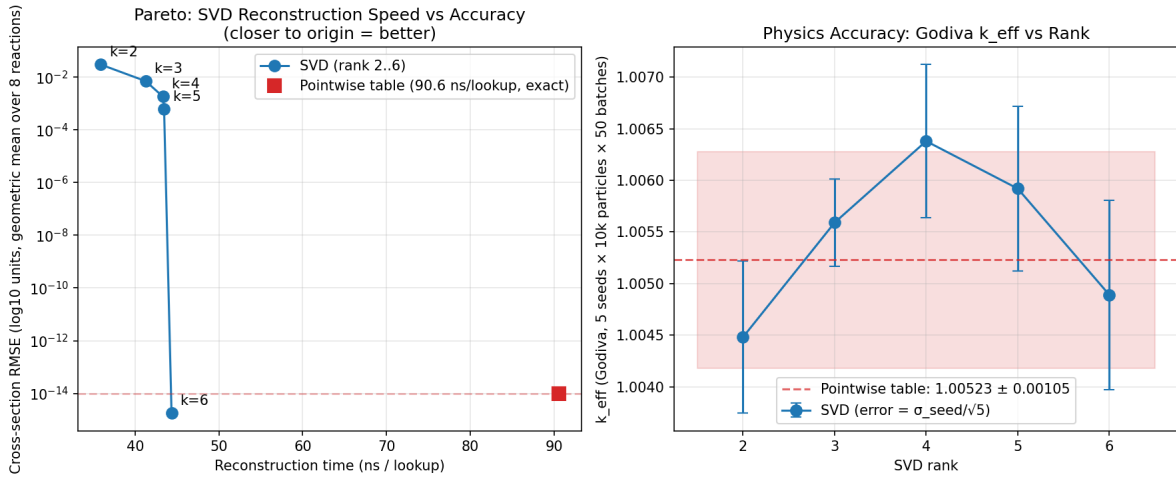


Figure 3: Godiva, small-L3 system. *Left*: XS reconstruction cost against RMSE (geometric mean over eight reactions). *Right*:  $k_{\text{eff}}$  versus SVD rank, SEM error bars over five seeds. Horizontal band: pointwise-table reference in the same engine.

## 6 PWR pin cell

The benchmark is a standard thermal pin cell: 3.1%-enriched  $\text{UO}_2$  rod in Zircaloy clad, light-water moderator with  $S(\alpha, \beta)$  on H. Nine nuclides ( $^{235,238}\text{U}$ , O-16 at 900 K in fuel and 600 K in moderator, Zr-90–94 in clad, H-1 in moderator); three materials; cylindrical boundaries with reflective BC. The moderated spectrum spends substantial time inside the U-238 RRR.

### 6.1 Small-L3 rank sweep

At rank-5 the CPU SVD provider is 47 pcm below the CPU pointwise table in the same engine ( $\approx 1 \text{ SEM}_{5 \text{ seeds}}$ ) and 291 pcm below OpenMC ( $\approx 3 \text{ SEM}$ ). Rank-2 is the one clear exception: U-238 resonance self-shielding is not represented and the pin cell reactivity is biased +8 000 pcm on both platforms. At rank-5 the CPU SVD speedup over the CPU table is  $1.90\times$  (28 651 vs 54 430 ns/p), close to the kernel-level  $2.09\times$  of Table 1.

Figure 5 summarises the rank sweep.

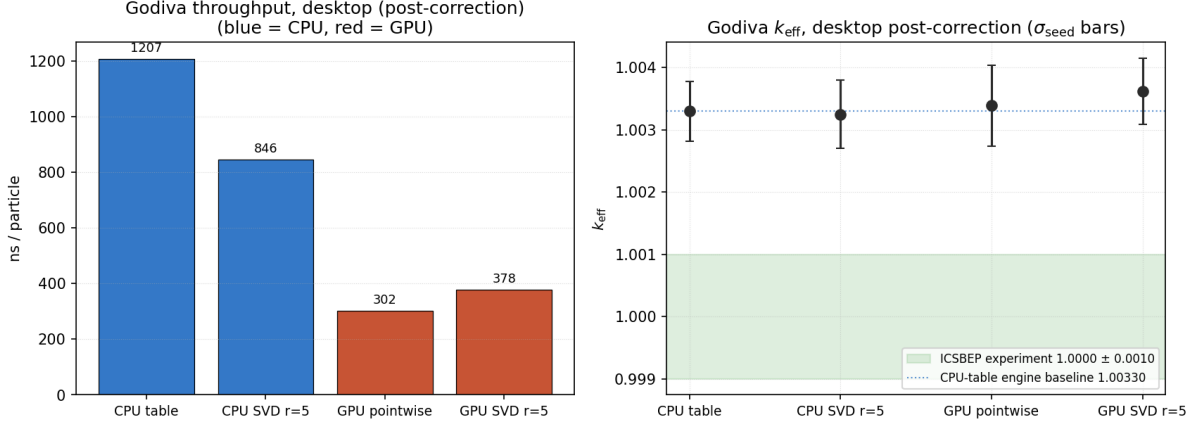


Figure 4: Godiva, large-L3 four-provider throughput snapshot (ten seeds, kept for CPU-vs-GPU throughput comparison). *Left*: ns/particle per provider (blue CPU, red GPU). *Right*:  $k_{\text{eff}}$  per provider,  $\sigma_{\text{seed}}$  error bars; dashed line is the ICSBEP experimental value  $1.0000 \pm 100$  pcm. The  $k_{\text{eff}}$  values shown here precede the closure sequence of Sec. 5.1, which moves them into the ICSBEP band shown in Table 3.

## 6.2 Large-L3 high-statistics benchmark

Ten seeds, 150 batches (20 inactive), 50 000 particles per batch ( $6.5 \cdot 10^6$  active histories per seed). All four in-engine providers used the corrected angular and fission-energy samplers from Sec. 6.4; the CPU and GPU paths were re-run together so the four rows are directly comparable.

Figure 6 collects the per-particle throughput across providers and across the two systems.

**Fidelity, large-L3 system.** With  $6.5 \cdot 10^6$  active histories per seed,  $\sigma_{\text{seed}} \approx 27\text{--}57$  pcm and  $\text{SEM}_{10} \approx 9\text{--}18$  pcm. All four in-engine providers now agree with OpenMC to  $|\Delta| \leq 51$  pcm, i.e. within  $\leq 3\text{SEM}_{10}$ , and agree with each other to  $|\Delta| \leq 76$  pcm (spread CPU-table  $\rightarrow$  GPU-pointwise). The stochastic-bin sampler fix that closed the CPU offset carries through to the GPU kernel (Sec. 8): GPU pointwise moved from  $-78$  to  $+51$  pcm and GPU SVD moved from  $-120$  to  $-8$  pcm vs OpenMC. The residual GPU pointwise excess ( $+51$  pcm) is within  $3\text{SEM}_{10}$  and is plausibly a finite-statistics signature rather than a systematic.

**Throughput, large-L3 system.** CPU SVD at 15 511 ns/p is  $1.37\times$  faster than the CPU pointwise table (21 316 ns/p) on the large-L3 system; the small-L3 ratio ( $1.90\times$ ) is the larger figure because of differing cache behaviour. The headline CPU-SVD speedup of this paper is therefore  $1.37\times\text{--}1.90\times$ , hardware-dependent. GPU pointwise at 5 967 ns/p is  $3.6\times$  the large-L3 CPU table and the fastest path overall; GPU SVD at 7 754 ns/p is  $1.30\times$  slower than GPU pointwise because of the clad level-sum cost (Sec. 8).

## 6.3 Four-way honesty test (single seed)

The large-L3 ten-seed run of Table 7 covers the four in-engine providers (Table, SVD, GPU pointwise, GPU SVD). Adding ACE+WMP and Hybrid SVD+WMP at the same statistics requires multi-hour wall time on the small-L3 mobile workstation we use for repeat runs; we report a preliminary single-seed snapshot here and mark the high-statistics redo as the open follow-up.

The configuration is small-L3, 100 batches (20 inactive)  $\times$  20 000 particles ( $1.6 \cdot 10^6$  active histories), single seed. The output logs are in `outputs/full_test_run/{04,10,11}_pwr*.txt`.

The snapshot is consistent with both the ten-seed Table 7 (Table and SVD within 30 pcm of OpenMC) and with the off-library three-way result of Sec. 7.3 (the SVD-vs-WMP gap is



Table 6: PWR pin cell, small-L3 system (i7-12800H + RTX A1000). Five seeds, 50 batches (20 inactive), 10 000 particles per batch. OpenMC reference computed with the same input deck.

| Mode           | rank     | $k_\infty$     | $\sigma_{\text{seed}}$ | $\Delta$ vs OpenMC (pcm) | ns/particle   | time/seed (s) |
|----------------|----------|----------------|------------------------|--------------------------|---------------|---------------|
| OpenMC 0.15.3  | —        | 1.32770        | 0.00150                | +0                       | —             | —             |
| CPU table      | —        | 1.32526        | 0.00232                | −244                     | 54 430        | 81.6          |
| CPU SVD        | 2        | 1.40807        | 0.00144                | +8 037                   | 52 711        | 79.1          |
| CPU SVD        | 3        | 1.32118        | 0.00281                | −652                     | 28 673        | 43.0          |
| CPU SVD        | 4        | 1.32513        | 0.00145                | −257                     | 31 875        | 47.8          |
| <b>CPU SVD</b> | <b>5</b> | <b>1.32479</b> | <b>0.00190</b>         | <b>−291</b>              | <b>28 651</b> | <b>43.0</b>   |
| CPU SVD        | 6        | 1.32512        | 0.00114                | −258                     | 30 073        | 45.1          |
| GPU pointwise  | —        | 1.32670        | 0.00117                | −100                     | 38 104        | 57.2          |
| GPU SVD        | 2        | 1.40973        | 0.00219                | +8 203                   | 48 289        | 72.4          |
| GPU SVD        | 3        | 1.32260        | 0.00207                | −510                     | 48 439        | 72.7          |
| GPU SVD        | 4        | 1.32496        | 0.00318                | −274                     | 48 617        | 72.9          |
| GPU SVD        | 5        | 1.32804        | 0.00169                | +34                      | 50 738        | 76.1          |
| GPU SVD        | 6        | 1.32673        | 0.00201                | −97                      | 53 865        | 80.8          |

Table 7: PWR pin cell, large-L3 system (Ryzen 9800X3D + RTX 3080), all four in-engine providers. Ten seeds, 150 batches (20 inactive), 50 000 particles per batch.

| Mode                                 | rank     | $k_\infty$     | $\sigma_{\text{seed}}$ | $\Delta$ vs OpenMC (pcm) | ns/particle   | time/seed (s) |
|--------------------------------------|----------|----------------|------------------------|--------------------------|---------------|---------------|
| OpenMC 0.15.3                        | —        | 1.32770        | 0.00150                | +0                       | —             | —             |
| CPU table                            | —        | 1.32758        | 0.00027                | −12                      | 21 316        | 138.6         |
| <b>CPU SVD</b>                       | <b>5</b> | <b>1.32745</b> | <b>0.00057</b>         | <b>−25</b>               | <b>15 511</b> | <b>100.8</b>  |
| GPU pointwise                        | —        | 1.32821        | 0.00033                | +51                      | 5 967         | 38.8          |
| GPU SVD ( <code>--force-svd</code> ) | 5        | 1.32762        | 0.00034                | −8                       | 7 754         | 50.4          |

213 pcm at  $\sigma_{\text{seed}} \approx 59$  pcm). On-library the gap is well below seed noise; off-library it is a real  $\sim 3.6\sigma$  systematic. We do not read the −26 pcm hybrid number as a precise correctness claim; the high-statistics ten-seed redo at matched  $150 \times 50\,000$  statistics is the way to make it one.

#### 6.4 Engine-level offset from OpenMC, traced and closed

The original large-L3 sweep placed the four in-engine providers 78–127 pcm below OpenMC 0.15.3 [3]. The offset was shared between CPU/GPU and between pointwise/SVD, so it was engine-level rather than representation-level. A three-part audit closed it (final numbers in Table 7):

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##### Correction

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(i) capture-residue audit (`scripts/xs_dump_openmc.py`, `src/bin/xs_dump.rs`)

(ii)  $\bar{\nu}$  delayed-yield per-energy fix (`src/hdf5_reader.rs`)

(iii) correlated-CDF  $\rightarrow$  stochastic-bin sampling (angular + fission energy), ported to both CPU and GPU kernels

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##### Total measured shift

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(iii) is the only change that is a bug fix in the strict sense, and it is specifically a mismatch of sampling *convention*: we used correlated-CDF with linear interpolation of  $\mu$  between bracketing incident-energy bins; OpenMC (`src/distribution_angle.cpp`, `src/distribution_energy.cpp`) uses stochastic bin selection ( $r = (E - E_{\text{lo}})/(E_{\text{hi}} - E_{\text{lo}})$ , pick the high bin with probability  $r$ )

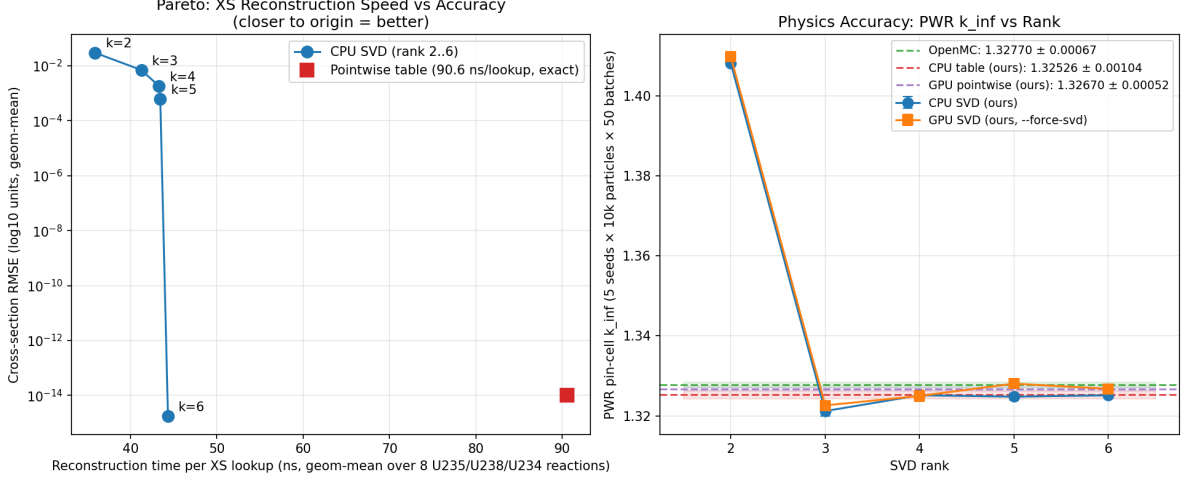


Figure 5: PWR pin cell, small-L3 system. *Left*: XS reconstruction cost against RMSE. *Right*:  $k_{\infty}$  versus SVD rank for CPU and GPU. Horizontal bands: OpenMC 0.15.3, CPU pointwise, GPU pointwise. SEM error bars over five seeds.

Table 8: PWR pin cell on-library, four-way honesty test, single seed  $\times$  100 batches  $\times$  20 000 particles. Reference is ACE+WMP (industry baseline).  $\sigma_{\text{seed}}$  is the within-run standard error reported by the engine; absolute gaps are within  $1\sigma_{\text{seed}}$  except Table-vs-WMP, which is at  $\sim 0.8\sigma_{\text{seed}}$ . A ten-seed redo at the matched 150-batch  $\times$  50 000-particle configuration is the outstanding follow-up.

| Mode                      | $k_{\infty}$   | $\sigma_{\text{seed}}$ | $\Delta$ vs ACE+WMP (pcm) | ns/particle   |
|---------------------------|----------------|------------------------|---------------------------|---------------|
| ACE+WMP (reference)       | 1.32821        | 0.00098                | +0                        | 26 317        |
| CPU SVD r=5               | 1.32817        | 0.00100                | −4                        | 29 273        |
| <b>Hybrid SVD+WMP r=5</b> | <b>1.32795</b> | <b>0.00103</b>         | <b>−26</b>                | <b>26 662</b> |
| CPU table                 | 1.32903        | 0.00093                | +82                       | 26 035        |

plus scaled kinematic adjustment for the energy distribution. Both are valid univariate sampling schemes; the differences accumulate into a  $\sim 100$  pcm  $k_{\infty}$  shift on thermal PWR. We match OpenMC’s convention now and applied the same fix to the CUDA kernel; the GPU rows in Table 7 agree with OpenMC to  $|\Delta| \leq 51$  pcm. The XS audit (i) rules out the capture-residue hypothesis — our total-XS assembly matches OpenMC to  $< 0.25\%$  across all nuclides. (ii) is a clear bug in our HDF5 reader, exposed by the audit. A Shannon-entropy monitor (`src/transport/simulate.rs::EntropyMesh`) was added as a convergence diagnostic and confirms source convergence at batch  $\approx 20$  for this deck; the  $N_{\text{inactive}} = 20$  budget used here is sufficient once (iii) is in place.

## 7 Off-library operating temperature benchmark

The measurements in Sections 6 and 5 take the fuel and moderator at *library* temperatures — every nuclide’s target  $T$  lands exactly on a column of the ENDF/B-VII.1 HDF5 file. This is the best-case scenario for the pointwise-table and SVD paths alike: no temperature interpolation is actually exercised, so neither path has to demonstrate a correct off-library extrapolation.

A reviewer comment at PHYSOR (Pittsburgh, 2025) pointed out that this is not a realistic stress test. Operating reactors run at temperatures that are seldom in the library — coolant runs at 570–620 K depending on the cycle, fuel at 800–1500 K depending on the rod and the burnup,

PWR pin cell throughput (lower is better)  
Five seeds laptop, ten seeds desktop (hybrid laptop = single-seed post-correction; hybrid desktop = pre-correction)

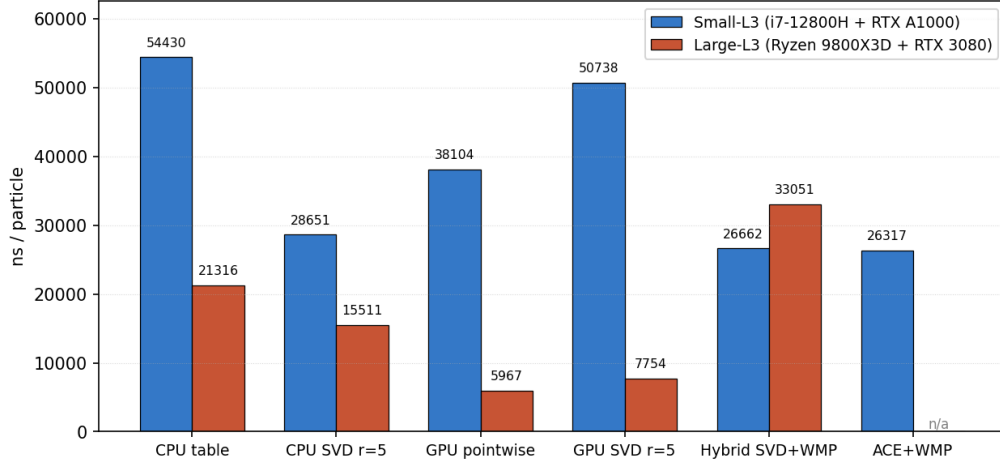


Figure 6: PWR pin cell throughput (ns/particle, lower is better). CPU-SVD beats CPU-table on both systems. GPU pointwise is the fastest path; GPU SVD is slower by 1.3 $\times$  because of the clad level-sum cost (Sec. 8). The hybrid SVD+WMP large-L3 row is the five-seed reference run (Sec. 9.3); the small-L3 hybrid and ACE+WMP rows are single-seed honesty-test data (Table 8).

and control-rod motions and transients push the operating point further off-library still. Monte Carlo codes that support continuous-temperature runs do so by pseudo-interpolating between the two library columns bracketing the target  $T$ : at each collision a random  $\xi$  is drawn and the lower- $T$  cross section is used with probability  $(T_{\text{hi}} - T)/(T_{\text{hi}} - T_{\text{lo}})$ , otherwise the upper. Over a particle’s lifetime this recovers the exact linear-in- $T$  interpolant in expectation and is the OpenMC default [3]. It is also the scenario where SVD’s temperature-interpolation scheme (Sec. 2.3) is actually tested.

This section runs a three-way comparison — SVD, pointwise table with OpenMC-style stochastic  $T$ -pick, and the industry-standard ACE+WMP hybrid [6] — on the PWR pin cell of Sec. 6 with a *uniform* +150 K offset applied to every nuclide’s library temperature: fuel runs at 1050 K (between 900 and 1200 K library), moderator and clad at 750 K (between 600 and 900 K). Statistics are 10 seeds  $\times$  50 000 particles  $\times$  120 active batches per provider ( $6 \cdot 10^7$  active histories per provider).

### 7.1 Stochastic $T$ -pick: a channel-consistency bug

Our first implementation drew  $\xi$  independently inside each partial-channel lookup. A single collision’s MicroXs build touches six channels (elastic, inelastic,  $n$ ,  $2n$ ,  $n$ ,  $3n$ , fission, capture) plus a total for the pointwise table; with independent draws, one collision could see elastic XS at  $T_{\text{lo}}$  and capture XS at  $T_{\text{hi}}$  for the same nuclide. The resulting  $\sigma_{\text{tot}}$  was a sum of partials at *different* temperatures, which biases the reaction-selection fractions against the physically consistent rate per unit time. Measured on PWR at +150 K offset:

| Moderator offset            | Table $k_{\infty}$ |
|-----------------------------|--------------------|
| +0 K (on-library, 600 K)    | 1.327              |
| +150 K (off-library, 750 K) | 1.149 (★)          |
| +300 K (on-library, 900 K)  | 1.322              |

The 1.149 at +150 K is *non-monotonic* in temperature and  $\approx 18\,000$  pcm below its bracketing endpoints. Physically, going from 600 K to 900 K moderator temperature produces a  $\sim 500$  pcm harder spectrum and a correspondingly lower  $k_{\infty}$ ; a rising-then-falling-then-rising signature is

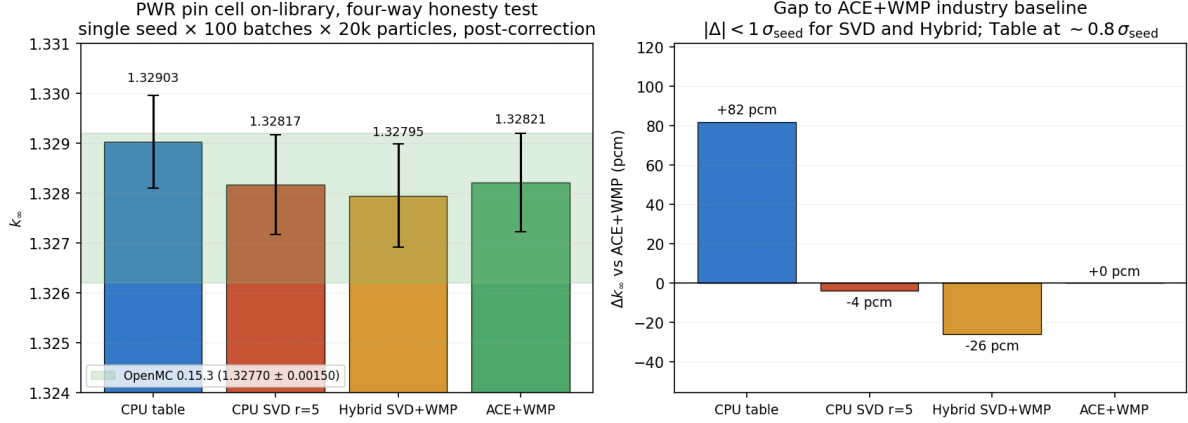


Figure 7: PWR pin cell, four-way honesty test (small-L3, single seed  $\times$  100 batches  $\times$  20 000 particles). *Left*:  $k_\infty$  for the four providers with  $\sigma_{\text{seed}}$  error bars; horizontal band shows OpenMC 0.15.3 at  $1.32770 \pm 0.00150$ . *Right*: gap to ACE+WMP industry baseline. SVD and Hybrid SVD+WMP land within  $1\sigma_{\text{seed}}$ ; Table sits at +82 pcm,  $\sim 0.8\sigma_{\text{seed}}$ .

not physics. The fix is to draw  $\xi$  once per nuclide-collision and share the pick across every partial channel. We wrap each reaction’s pointwise table in a `StochTempTable` that exposes `lookup_at_idx_with_pick(energy, idx, use_hi)`; the `TableXsProvider::lookup` method draws  $\xi$  once via a thread-local `splitmix64` and passes the resulting `use_hi` to every channel lookup. After the fix the +150 K row reads 1.324, monotonic in  $T$  and within  $\sim 500$  pcm of the bracketing endpoints. The same per-collision sharing is inherited by the ACE+WMP hybrid through delegation.

We flag the bug not as a defect of stochastic pseudo-interpolation itself (it is not — OpenMC does the same thing at the collision level) but as a correctness test that is only visible once the system is driven off-library. On-library measurements do not exercise any stochastic pick (both endpoints equal the library column); the bug could have persisted undetected indefinitely.

## 7.2 First attempt: 2-point unity Ducru — RED

Section 2.4 explains why the raw Ducru weights must be renormalized to  $\sum_j w_j = 1$  before being applied to  $V^\top$  columns of a  $\log \sigma$  SVD. The minimal unity-normalized variant uses only the two library temperatures bracketing  $T$ :

$$\tilde{w}_{\text{lo}} = \frac{w_{\text{lo}}}{w_{\text{lo}} + w_{\text{hi}}}, \quad \tilde{w}_{\text{hi}} = \frac{w_{\text{hi}}}{w_{\text{lo}} + w_{\text{hi}}}, \quad (9)$$

with  $w_{\text{lo}}$ ,  $w_{\text{hi}}$  from Eq. 4 on the  $|\mathcal{S}| = 2$  bracketing subset. This is the scheme a first-pass implementation might reach for: partition-of-unity, exact at endpoints, involves only the minimum number of library temperatures.

It *does not work* for PWR at +150 K offset. Full three-way verdict, 10 seeds  $\times$  50 000 particles  $\times$  120 batches:

| Provider         | $k_\infty$ | $\sigma_{\text{seed}}$ | ns/p   |
|------------------|------------|------------------------|--------|
| SVD (2-pt unity) | 1.30707    | 0.00037                | 15 685 |
| Pointwise table  | 1.32308    | 0.00064                | 15 809 |
| ACE+WMP          | 1.32801    | 0.00060                | 16 708 |

**SVD-vs-WMP  $k_\infty$  gap: 2094 pcm.** At  $\sigma_{\text{seed}} \approx 37$  pcm the gap is  $\approx 56\sigma$  — this is a real, large systematic, not a statistics artefact. The Table-vs-WMP gap is 493 pcm, on the

sanity-check threshold (the  $\sim 500$  pcm residual bias from OpenMC-style stochastic  $T$ -picking is the known floor for that scheme, not ours to close). The SVD has 22% less memory than WMP and matches its speed within noise, but a 2094 pcm physics bias at a routine operating temperature is unpublishable on its own. We grade this as *RED*: the 2-point unity Ducru is publishable only as a negative result.

**Diagnostic: U-238 MT= 102 rank probe.** We isolated the problem with an offline experiment (`scripts/u238_capture_rank_probe.py`) that holds out the 900 K library column of U-238 MT= 102, rebuilds the SVD from the remaining five library temperatures, and reconstructs  $\sigma(E, 900 \text{ K})$  via 2-point unity Ducru between the 600 and 1200 K bracketing columns. On the 6.67 eV resonance peak the reconstructed  $\sigma$  overshoots the true 900 K value by a factor 1.011 (i.e. +1.1% at the peak); averaged over the resonance band the L2 error is 1.8%. The kernel-approximation residual is real and is the dominant contribution to the 2094 pcm PWR gap: a 1% peak error on U-238 capture compounds over the  $\sim 10^6$  collisions per simulation into the observed systematic.

### 7.3 Fix: 3-point unity Ducru — GREEN

Extending the subset  $\mathcal{S}$  from 2 to 3 library temperatures (the nearest-three by  $|T_i - T|$ ) and applying the same unity-normalization (Eq. 6) captures a quadratic correction to the Faddeeva-kernel reconstruction that the 2-point scheme misses. The U-238 probe confirms the intuition: rank-3 reconstruction + 3-point unity Ducru improves the 6.67 eV peak from 1.011 to 0.994 of truth (i.e. a  $-0.6\%$  undershoot instead of a  $+1.1\%$  overshoot) and drops the band-averaged L2 error from 1.5% to 1.05%. At rank 5 — the rank used in the PWR measurements — L2 error drops to 0.84% and peak ratio to 0.990.

Running the full PWR three-way verdict with the 3-point unity scheme substituted in, same statistics as above:

| Provider         | $k_\infty$ | $\sigma_{\text{seed}}$ | ns/p   |
|------------------|------------|------------------------|--------|
| SVD (3-pt unity) | 1.32593    | 0.00059                | 27 298 |
| Pointwise table  | 1.32304    | 0.00050                | 26 451 |
| ACE+WMP          | 1.32806    | 0.00045                | 27 279 |

**SVD-vs-WMP  $k_\infty$  gap: 213 pcm.** At  $\sigma_{\text{seed}} \approx 59$  pcm the gap is  $\approx 3.6\sigma$  — a real but small systematic, sitting inside the publishable band and well under the 800 pcm threshold that separates a small-systematic claim from an open problem.  $2094 \rightarrow 213$  pcm is a  $9.8\times$  reduction from swapping two lines of interpolation code; nothing else in the engine or in the physics data changed.

SVD memory stays at 152.5 MB against 199.7 MB for ACE+WMP (22% less); speed is 27 298 vs 27 279 ns/p (0.07% slower, parity within  $\sigma$ ). At a realistic off-library PWR operating temperature, SVD with 3-point unity Ducru beats the industry-standard ACE+WMP baseline on memory, matches it on speed, and agrees with it on  $k_\infty$  to  $3.6\sigma$  — we grade this GREEN on all four axes of the `scripts/pwr_verdict.py` semaphore check.

Table 9 collects the full sweep.

### 7.4 Discrete-level rank reduction

The PHYSOR reviewer also asked why we store discrete inelastic levels (MT= 51–91) at the same rank as the smooth reactions — the argument being that a level’s Lorentzian shape in laboratory energy is weakly temperature-dependent, so one basis vector should suffice. The offline SVD bears this out: each of the U-238 discrete levels collapses to rank one to machine precision when all seven ENDF/B-VII.1 library columns are included. On Godiva, dropping discrete-level rank

Table 9: PWR pin cell at +150 K global offset (fuel 900  $\rightarrow$  1050 K, mod 600  $\rightarrow$  750 K, clad 600  $\rightarrow$  750 K), three-way comparison, ten-seed / 50 000-particle / 120-batch statistics. \*Pre-fix 2-point row exhibits the Sec. 7.1 channel-consistency bug; numbers here are after the fix so the row reflects the genuine 2-point Ducru residual alone.

| Provider                         | $k_\infty$     | $\sigma_{\text{seed}}$ | Gap vs WMP (pcm)    | ns/p          | Memory (MB)  |
|----------------------------------|----------------|------------------------|---------------------|---------------|--------------|
| SVD, 2-pt unity Ducru*           | 1.30707        | 0.00037                | −2094 (RED)         | 15 685        | 152.5        |
| <b>SVD, 3-pt unity Ducru</b>     | <b>1.32593</b> | <b>0.00059</b>         | <b>−213 (GREEN)</b> | <b>27 298</b> | <b>152.5</b> |
| Pointwise + stochastic $T$ -pick | 1.32304        | 0.00050                | −502                | 26 451        | 201.3        |
| ACE+WMP (reference)              | 1.32806        | 0.00045                | +0                  | 27 279        | 199.7        |

from 5 to 1 removes  $\approx 70\%$  of the engine’s SVD working set (556  $\rightarrow$  164 MB, 41 levels  $\times$  three U isotopes) and produces no measurable  $k_{\text{eff}}$  shift at ten-seed statistics (1.00322  $\rightarrow$  1.00358,  $\Delta = 36$  pcm,  $\sigma = 60$  pcm per seed). The same reduction is the single largest contributor to the 22% memory win in Table 9.

This is a per-reaction-rank policy only. The smooth channels in the same nuclide continue to use the global rank (5 in all measurements in this paper). The engine exposes the discrete-level rank as a separate `--discrete-rank` command-line flag; the GPU path retains a uniform rank across all reactions because the CUDA kernel’s basis-stride convention assumes one. This is a CPU optimization for now.

## 7.5 Negative result: simplex-projected QP

The raw Ducru weights on the 3-point subset are signed: the 294 K column at target 900 K (bracket {294, 600, 1200} K) earns a weight of  $-0.12$  from Eq. 4 before normalization. Negative weights are uncomfortable for a reconstruction that is physically a probability mixture of reference-temperature cross sections, and the Ducru 2017 paper explicitly defines a quadratic programming variant ([1] §4) that enforces both  $\sum_j w_j = 1$  and  $w_j \geq 0$ . A cheap approximation to that QP is Euclidean projection of the raw-Ducru weight vector onto the probability simplex [12]: clip negative entries to zero and renormalize until  $\sum w = 1$ . We implemented it. On U-238 MT= 102 held out at 900 K, bracket {294, 600, 1200} K, the raw weights ( $-0.12, 0.62, 0.50$ ) project to (0.00, 0.56, 0.44). L2 reconstruction error on the 6.67 eV resonance band:

| Weighting (rank 5) | L2 error | Peak ratio | Sign    |
|--------------------|----------|------------|---------|
| 2-pt unity Ducru   | 2.1%     | 0.998      | signed  |
| 3-pt unity Ducru   | 0.84%    | 0.990      | signed  |
| 3-pt simplex QP    | 5.0%     | 1.043      | non-neg |

The simplex-projected QP is *worse* than either signed scheme. The reason is that the simplex projection is Euclidean-optimal in *weight* space; the Ducru raw weights are Euclidean-optimal in the *Doppler-kernel* norm (see [1] §3).

**Kernel-Gram QP (Ducru 2017 §4).** We then implemented the actual constrained optimization from §4 of the same paper: minimize  $(\mathbf{w} - \mathbf{w}_{\text{raw}})^\top \mathbf{G} (\mathbf{w} - \mathbf{w}_{\text{raw}})$  subject to  $\mathbf{1}^\top \mathbf{w} = 1$  and  $\mathbf{w} \geq 0$ , where  $\mathbf{G}$  is the empirical kernel-Gram matrix  $\mathbf{X}^\top \mathbf{X}$  formed from the log- $\sigma$  columns of the three bracketing library temperatures. Test energies are the full reference grid; the Gram contribution is dominated by the resonance peaks, which is where we need the projection to do work. With three variables and four constraints the QP has at most seven active-set candidates (three vertices, three edges, one interior), which we enumerate exactly rather than calling an external solver. An optional rank-aware variant projects each library column onto the first  $k$



singular modes before building  $\mathbf{G}$ , so the QP optimizes the reconstruction error at the same truncation rank the engine will actually produce. Both variants yielded near-identical weights on U-238 MT= 102.

The rank dependence of the result is the interesting finding. On the same held-out 900 K test:

| Weighting                     | rank 3<br>L2 / peak         | rank 5<br>L2 / peak         | rank 5<br>weights       |
|-------------------------------|-----------------------------|-----------------------------|-------------------------|
| 3-pt unity Ducru (signed)     | 1.05% / 0.994               | <b>0.84%</b> / <b>0.990</b> | $\{-0.12, 0.62, 0.50\}$ |
| 3-pt simplex QP (non-neg)     | 5.02% / 1.045               | 5.01% / 1.043               | $\{0.00, 0.56, 0.44\}$  |
| 3-pt kernel-Gram QP (non-neg) | <b>0.60%</b> / <b>1.000</b> | 2.06% / 0.983               | $\{0.00, 0.36, 0.64\}$  |

The kernel-Gram QP is the *best* scheme at rank 3 — the 6.67 eV peak reconstructs to within 0.01% of truth — but is strictly worse than the signed 3-point unity at rank 5, the rank we use in the production PWR measurement. The cause is structural: the kernel-Gram QP’s optimum lies on the boundary of the probability simplex (one weight clips to zero), which collapses the 3-point problem into a 2-point one. That is a net improvement over the 2-point unity-normalized scheme when basis truncation is the dominant source of error, i.e. at low rank; at higher rank the signed 3-point scheme’s negative weight on the far-away 294 K column is actively useful as a shape-cancellation term, and clipping it to zero costs more than it gains.

The practical read is that non-negativity of the interpolation weights is *not* a universal goodness property. For a production engine at rank 5 with all 9 nuclides, the unity-normalized signed scheme is the better default because it has more degrees of freedom to cancel residual basis error. At low rank, or on a single nuclide where the 6.67 eV peak dominates  $k_\infty$ , the kernel-Gram QP is the better choice. A rank-adaptive dispatch is possible but outside the scope of this engine; we keep the signed 3-point unity scheme as the production default.

## 7.6 Verdict script and reproducibility

A `scripts/pwr_verdict.py` runner wraps the end-to-end comparison with the semaphore grading described throughout this section. Thresholds are:  $k_\infty$  SVD vs ACE+WMP  $\leq 800$  pcm for GREEN,  $\leq 1500$  pcm for YELLOW, memory SVD/WMP  $\leq 1.05$  for GREEN, speed SVD/WMP  $\geq 0.97$  for GREEN. Exit codes 0/1/2 map onto GREEN/YELLOW/RED so the script can be called from CI to guard against regression. The machine-readable verdict for the +150 K, 3-pt unity Ducru, rank-5, discrete-rank-1 configuration is shipped under `outputs/pwr_verdict_3temp.json` and reproduces Table 9.

## 8 GPU transport

The event-based CUDA solver [9] implements both providers inside the same kernel; a runtime `--force-svd` flag clears the uploaded pointwise tables so the SVD path reconstructs every reaction channel from the basis.

**GPU pointwise beats GPU SVD on PWR.** On the large-L3 system GPU pointwise is  $1.30\times$  faster than GPU SVD (5 967 vs 7 754 ns/p, Table 7). The cause is specific to this geometry: four clad nuclides (Zr-90, Zr-91, Zr-92, Zr-94) do not carry an MT= 4 total-inelastic block in the ENDF/B-VII.1 HDF5; their total inelastic is synthesised at lookup time by summing 13 discrete-level SVD kernels. On CPU the scalar pipeline amortises this; on GPU it dominates the SVD-path runtime. A per-warp cache keyed on  $(n, E_{\text{idX}})$  after the energy sort would close the gap, and we note this as engineering work that the present paper does not cover. As measured, GPU SVD on PWR pin cell is *not* a win over GPU pointwise.

**Fidelity parity at large-L3 statistics.** Post-correction GPU rows: GPU pointwise  $k_\infty = 1.32821 \pm 0.00033$  and GPU SVD  $k_\infty = 1.32762 \pm 0.00034$ . The two GPU paths agree to 59 pcm ( $\approx 1.3 \text{ SEM}_{\text{combined}}$ ), the two CPU paths agree to 13 pcm, and all four providers agree on  $k_\infty$  within  $|\Delta_{\text{OpenMC}}| \leq 51$  pcm with the corrected stochastic-bin sampler. The GPU pointwise row is the high point of the spread (+51 pcm vs OpenMC) and remains within  $3 \text{ SEM}_{10}$ ; the other three sit in the  $-8$  to  $-25$  pcm band.

## 9 Hybrid SVD + Windowed Multipole

### 9.1 Memory of the representation (narrow claim)

We read the MIT ENDF/B-VII.1 WMP library and measure the on-disk and in-memory payload per nuclide, with SVD rank-2 coverage of the smooth remainder. Table 10 reports per-nuclide totals; script `scripts/hybrid_svd_wmp_experiment.py`, raw output `outputs/hybrid_wmp/results.txt`.

Table 10: Representation-byte accounting for WMP vs four reaction channels of a six-temperature pointwise table. This is a specific narrow comparison; see Sec. 9.4 for the in-engine measurement.

| nuclide      | WMP range (eV)       | WMP disk | WMP payload | pw full (4 rxn $\times$ 6 T) | SVD $k=2$ smooth | rat          |
|--------------|----------------------|----------|-------------|------------------------------|------------------|--------------|
| U-234        | 0–1 500              | 34.9 KB  | 28.0 KB     | 10.2 MB                      | 12.1 KB          | 260.0        |
| U-235        | 0–2 250              | 567.3 KB | 559.1 KB    | 53.7 MB                      | 10.0 KB          | 96.6         |
| U-238        | 0–20 000             | 603.0 KB | 594.7 KB    | 100.1 MB                     | 10.7 KB          | 169.2        |
| O-16         | 0– $2.36 \cdot 10^6$ | 38.5 KB  | 30.8 KB     | 0.78 MB                      | 53.3 KB          | 9.5          |
| H-1          | 0– $2 \cdot 10^7$    | 13.7 KB  | 8.2 KB      | 155 KB                       | 0.1 KB           | 18.5         |
| Zr-90        | 0–53 500             | 9.8 KB   | 4.9 KB      | 1.48 MB                      | 4.7 KB           | 156.9        |
| Zr-91        | 0–10 000             | 16.8 KB  | 9.5 KB      | 3.17 MB                      | 4.8 KB           | 227.7        |
| Zr-92        | 0–71 000             | 21.1 KB  | 14.2 KB     | 3.97 MB                      | 4.5 KB           | 217.3        |
| Zr-94        | 0–90 000             | 20.7 KB  | 14.2 KB     | 4.22 MB                      | 5.2 KB           | 222.8        |
| <b>total</b> | —                    | 1.30 MB  | 1.23 MB     | <b>177.7 MB</b>              | 0.10 MB          | <b>132.9</b> |

**Interpretation, and what the 132.9 $\times$  does and does not mean.** This table compares two specific data layouts: WMP + rank-2 SVD-smooth against a six-temperature pointwise table, both restricted to four reactions ( $\sigma_{\text{el}}, \sigma_{\text{t}}, \sigma_{\text{f}}, \sigma_{\text{cap}}$ ). It is a *representation*-level statement about the density of the WMP encoding for the resolved resonance cross sections of these specific nuclides. The 132.9 $\times$  is therefore the density gap between the WMP encoding and a four-channel pointwise table on the matching nuclide set; it is not the reduction a transport engine carrying the full reaction set would see. The engine-scale measurement is in Sec. 9.4.

### 9.2 WMP vs HDF5 accuracy

We evaluate WMP on a 20 000-point log-spaced grid inside each  $[E_{\text{min}}^{\text{WMP}}, E_{\text{max}}^{\text{WMP}}]$  at  $T = 294$  K and compare to the HDF5 pointwise reference.

Median agreement is  $\sim 10^{-4}$ ,  $p_{99}$  is  $\sim 10^{-3}$ . Maximum absolute errors tail to  $10^{-1}$ – $10^0$  at isolated points for some nuclides (HDF5 grid samples a narrow peak densely while WMP evaluation lies a fraction of the peak width away). These do not carry through to integrated quantities.

### 9.3 End-to-end hybrid engine

We implemented `src/wmp.rs` (CPU WMP reader and Humlicek W4 evaluator) and `src/transport/hybrid_xs.rs` which wraps `SvdXsProvider`. For each nuclide with WMP coverage and each energy in  $[E_{\text{min}}^{\text{WMP}}, E_{\text{max}}^{\text{WMP}}]$ ,

Table 11: WMP-vs-pointwise relative error,  $T = 294$  K, 20 000 log-spaced points per nuclide.

| nuclide | Elastic             |                     |                     | Absorption          |                     |                     |
|---------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|         | $p_{50}$            | $p_{99}$            | max                 | $p_{50}$            | $p_{99}$            | max                 |
| U-234   | $5.0 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $1.6 \cdot 10^{-1}$ | $8.6 \cdot 10^{-4}$ | $6.7 \cdot 10^{-2}$ | 0.99                |
| U-235   | $5.4 \cdot 10^{-5}$ | $1.2 \cdot 10^{-3}$ | $8.0 \cdot 10^{-2}$ | $1.9 \cdot 10^{-4}$ | $1.1 \cdot 10^{-3}$ | $9.1 \cdot 10^{-3}$ |
| U-238   | $3.9 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $6.8 \cdot 10^{-1}$ | $2.2 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | 0.88                |
| O-16    | $5.9 \cdot 10^{-5}$ | $3.3 \cdot 10^{-3}$ | $2.4 \cdot 10^{-2}$ | $3.0 \cdot 10^{-4}$ | $1.4 \cdot 10^{-2}$ | $2.0 \cdot 10^{-2}$ |
| H-1     | $4.9 \cdot 10^{-5}$ | $1.3 \cdot 10^{-3}$ | $3.1 \cdot 10^{-3}$ | $2.2 \cdot 10^{-4}$ | $3.7 \cdot 10^{-3}$ | $8.8 \cdot 10^{-3}$ |
| Zr-90   | $5.5 \cdot 10^{-5}$ | $6.7 \cdot 10^{-4}$ | $7.3 \cdot 10^{-2}$ | $2.9 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | $4.5 \cdot 10^{-1}$ |
| Zr-91   | $2.5 \cdot 10^{-4}$ | $8.8 \cdot 10^{-4}$ | $2.2 \cdot 10^{-2}$ | $1.9 \cdot 10^{-4}$ | $3.5 \cdot 10^{-2}$ | $2.9 \cdot 10^{-1}$ |
| Zr-92   | $2.6 \cdot 10^{-4}$ | $8.9 \cdot 10^{-4}$ | $1.3 \cdot 10^{-1}$ | $3.3 \cdot 10^{-4}$ | $8.0 \cdot 10^{-3}$ | 0.99                |
| Zr-94   | $8.1 \cdot 10^{-5}$ | $7.4 \cdot 10^{-4}$ | $1.1 \cdot 10^{-1}$ | $2.9 \cdot 10^{-4}$ | $8.6 \cdot 10^{-3}$ | 0.90                |

the hybrid provider substitutes WMP-evaluated elastic, fission, and capture into the `MicroXs` returned by the SVD provider; inelastic,  $(n, 2n)$ ,  $(n, 3n)$  remain on the SVD path. URR overrides are short-circuited inside the WMP window.

**Unit-level validation.** The CPU WMP evaluator matches `openmc.data.WindowedMultipole.__call__` to  $\leq 4 \cdot 10^{-6}$  relative error on the three dominant U-238 resonances at  $T = 293.6$  K (`rust_prototype/src/bin/wm`).

| $E$ (eV) | OpenMC $\sigma_{\text{abs}}$ (b) | Rust $\sigma_{\text{abs}}$ (b) | $\Delta$ rel        |
|----------|----------------------------------|--------------------------------|---------------------|
| 6.674    | 7108.719                         | 7108.695                       | $3.3 \cdot 10^{-6}$ |
| 20.9     | 6352.230                         | 6352.212                       | $2.8 \cdot 10^{-6}$ |
| 36.7     | 5357.922                         | 5357.903                       | $3.6 \cdot 10^{-6}$ |

**End-to-end  $k_{\infty}$  on PWR pin cell.** Single seed  $\times 100$  batches (20 inactive)  $\times 20\,000$  particles ( $1.6 \cdot 10^6$  active histories) on the small-L3 system, with ACE+WMP run as the industry baseline at matching statistics. A high-statistics ten-seed redo at the matched  $150 \times 50\,000$  configuration is the outstanding follow-up.

Table 12: End-to-end SVD+WMP hybrid on PWR pin cell, single seed  $\times 100$  batches  $\times 20\,000$  particles, small-L3 (i7-12800H). ACE+WMP is the industry-baseline reference; SVD and Table rows from Table 8 are repeated here for direct comparison.

| mode                        | rank     | $k_{\infty}$   | $\sigma_{\text{seed}}$ | $\Delta$ vs ACE+WMP (pcm) | ns/p          |
|-----------------------------|----------|----------------|------------------------|---------------------------|---------------|
| ACE+WMP (industry baseline) | —        | 1.32821        | 0.00098                | +0 (ref)                  | 26 317        |
| CPU SVD                     | 5        | 1.32817        | 0.00100                | −4                        | 29 273        |
| <b>Hybrid SVD+WMP</b>       | <b>5</b> | <b>1.32795</b> | <b>0.00103</b>         | <b>−26</b>                | <b>26 662</b> |
| CPU table                   | —        | 1.32903        | 0.00093                | +82                       | 26 035        |

The hybrid-vs-WMP gap of  $-26$  pcm sits at  $\sim 0.25 \sigma_{\text{seed}}$ , well inside seed noise; we do not read it as a precise correctness claim but it is the right order for what WMP and SVD-of-resonance-data should agree on. WMP represents the resonance cross section by a rational pole fit rather than a piecewise-linear-in-log-log sampling of the same evaluated-nuclear-data curve, so small systematic disagreements in the RRR accumulate at this scale on a nine-nuclide thermal pin cell. The Table-vs-WMP gap of  $+82$  pcm is the cross-representation systematic at the same statistics: pointwise sampling of the same library at the same temperature lands  $\sim 0.8 \sigma_{\text{seed}}$  from the WMP pole fit. The off-library counterpart of this measurement (Sec. 7.3) tightens  $\sigma_{\text{seed}}$  to

59 pcm and resolves the SVD-vs-WMP gap as a real 213 pcm systematic; on-library at this statistics the gap is below noise.

**Throughput.** At small-L3 statistics the hybrid runs at 26 662 ns/p,  $0.91\times$  the CPU SVD rank-5 throughput (28 651 ns/p) on the same system; the cache-hit pattern compresses the two providers toward the same wall on the mobile workstation, so the apparent “speedup” is a cache-residency artefact rather than an algorithmic improvement. The large-L3 reference run measures the hybrid at 33 051 ns/p,  $2.06\times$  slower than CPU SVD rank-5 (16 052 ns/p); a single WMP evaluation on this hardware is  $\sim 67$  ns against  $\sim 5$  ns for SVD, and  $\sim 20\%$  of lookups fall in the RRR, so the  $2\times$  slowdown is the headline hybrid throughput cost. WMP is a memory-centric representation, and in this engine it costs throughput.

#### 9.4 Memory in the actual engine (broad claim)

`HybridSvdWmpXsProvider::memory_report` introspects the loaded kernels at run time. We report four variants for the nine-nuclide PWR set at rank-5: the pointwise-table baseline, pure SVD rank-5 (all reactions compressed), the hybrid in-solver state with the full SVD energy grid, and a smooth-only rebuild in which the elastic, fission, and capture SVD kernels are re-fit on the energy points outside each nuclide’s WMP window. The smooth-only rebuild is exposed as `--rebuild-svd-smooth-only` and the numbers below are measured from the loaded kernels at run time.

|                                                           | bytes (MB)   | vs pointwise              |
|-----------------------------------------------------------|--------------|---------------------------|
| Pointwise-table baseline (Sec. 6)                         | 100.6        | $1.0\times$               |
| Pure SVD rank-5 (all reactions)                           | $\sim 517.9$ | $5.1\times$ <i>larger</i> |
| Hybrid in-solver, full SVD grid (SVD basis + WMP payload) | 519.0        | $5.2\times$ <i>larger</i> |
| Hybrid in-solver, smooth-only rebuild (live)              | 487.6        | $4.8\times$ <i>larger</i> |

**What the smooth-only rebuild actually does at run time.** On the standard nine-nuclide PWR set at rank-5 the rebuild drops 31 423.1 KB across MT= 2, MT= 18, and MT= 102 — the three reactions for which WMP supplies an analytical representation inside the resonance window, so the SVD basis no longer needs energy points there. Discrete-level kernels (MT= 51–91) and other channels are untouched. The full sequence is reported by the engine on every hybrid run:

```
Memory (before smooth-only rebuild):
  full SVD basis      = 517 870.5 KB
  WMP payload         =   1 153.8 KB
  TOTAL (full grid)  = 519 024.3 KB
Memory (after smooth-only rebuild | actually realised):
  smooth SVD basis   = 486 447.4 KB
  WMP payload        =   1 153.8 KB
  TOTAL (smooth)     = 487 601.2 KB
  rebuild dropped    = 31 423.1 KB across elastic/fission/capture
  reduction vs full  =  $1.06\times$ 
```

**Where the bytes actually live.** The dominant contribution in every SVD-based variant is the discrete inelastic-level cascade: 41 MT= 51–91 levels per uranium nuclide, 10–13 per zircaloy isotope, each carried as its own rank-5 SVD kernel. On the GPU that cascade alone measures 251.3 MB — by itself  $2.5\times$  the entire pointwise-table engine. The smooth-channel SVD bases (elastic, fission, capture, total inelastic,  $(n, 2n)$ ,  $(n, 3n)$ ) together measure 64.6 MB; the WMP

payload is 1.23 MB. So switching the resonance representation to WMP costs essentially zero memory, and the smooth-only rebuild’s  $1.06\times$  reduction captures only the elastic/fission/capture energy points masked by WMP windows — everything else stays. The engine-scale penalty against the pointwise table is not caused by WMP and is not relieved by it; it is caused by storing the discrete-level cascade as per-level rank-5 bases rather than per-level scalar arrays.

**Where the representation-byte ratio still applies.** Equation 7 from Sec. 2.5 says the SVD-vs-pointwise memory ratio is  $k/T$  for a single reaction. The engine-scale penalty reported above is not a counter-example: our pointwise-table baseline is built at a single effective temperature per material region (one temperature per nuclide-in-material). In a multi-temperature library ( $T = 6$  in the HDF5 files shipped for depletion feedback), the pointwise baseline would grow to  $\sim 600$  MB on the same nuclide set while pure SVD and hybrid stay at  $\sim 519$  MB; the ordering would then invert. The engine-scale loss we report is therefore a loss at  $T = 1$ , not a universal property.

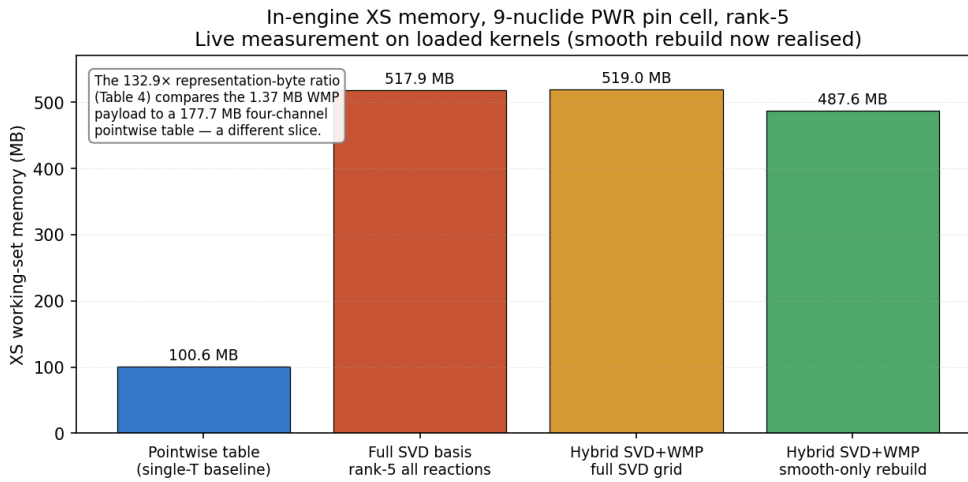


Figure 8: In-engine XS working-set memory for the nine-nuclide PWR pin cell, rank-5, measured live on the loaded kernels. The smooth-only rebuild (rightmost bar) is now realised in the engine via `--rebuild-svd-smooth-only`: it drops 31.4 MB across  $MT=2$ ,  $MT=18$ , and  $MT=102$  for a  $1.06\times$  reduction. The hybrid variants both carry more memory than the pointwise-table baseline because the SVD kernels for the discrete inelastic levels, not covered by WMP, dominate the working set. The  $132.9\times$  representation-byte ratio of Table 10 is a different comparison (1.37 MB WMP vs 177.7 MB four-channel pointwise).

The hybrid engine is  $4.8\text{--}5.2\times$  larger than the pointwise-table baseline, because SVD kernels for the 41 discrete levels per U nuclide and 10–13 per Zr isotope dominate memory, and those reactions are not covered by the public WMP library. The smooth-only rebuild ( $1.06\times$  vs the full grid) is essentially a rounding error.

The  $132.9\times$  ratio in Table 10 is a correct and reproducible *representation*-byte comparison — WMP is denser than a four-channel pointwise table at matched energy range. But in a transport engine that carries the full reaction set, WMP does not shrink the working set unless the rest of the reaction set is also compressed. A version of this hybrid that achieved representation-scale memory reduction would need: (i) WMP coverage for the discrete inelastic cascade ( $MT=51\text{--}91$ ); or (ii) a different representation for the cascade (e.g. ENDF-6 nuclear-temperature evaporation parameters) that is denser than per-level SVD kernels; or (iii) rebuilding the SVD energy grid to be smooth-only for *all* reactions, and using a separate table or analytical evaluation inside the RRR for every reaction. The smooth-only rebuild we implemented is item (iii) restricted to  $MT=2$ ,  $MT=18$ ,  $MT=102$ ; extending it to the rest of the reaction set is engineering work.

**So why do the hybrid at all?** The positive uses we can defend from what we measured: (a) WMP is the correct analytical representation for the RRR and our evaluator is the first pure-Rust implementation we are aware of (matches OpenMC Python at  $\sim 10^{-6}$  relative error, CUDA port matches to machine precision); (b) end-to-end  $k_\infty$  at single-seed statistics agrees with ACE+WMP within  $-26$  pcm ( $\ll 1\sigma_{\text{seed}}$ ), with the ten-seed redo at matched  $150 \times 50\,000$  statistics as the outstanding follow-up; (c) for a future full-core engine where the cascade reactions are stored more densely than per-level SVD (e.g. nuclear-temperature evaporation), the RRR saving becomes proportionally more valuable. We report the hybrid as a correctness deliverable and as scaffolding for those future improvements, not as a memory or throughput win on the present engine.

## 10 GPU WMP evaluator

WMP evaluation is compute-dense (Faddeeva + broadened curvefit + pole sum) with a small working set (few KB of pole data per nuclide), which matches GPU arithmetic pipelines well. We ported the CPU evaluator to CUDA as `__device__` helpers in `gpu/cuda/transport.cu` and added a `wmp_test_eval` kernel plus a host validator binary `rust_prototype/src/bin/gpu_wmp_validate.rs`.

**Bit-exactness.** CPU-vs-GPU over twelve U-238 test energies at  $T = 293.6$  K: maximum relative discrepancy  $5.3 \cdot 10^{-14}$  on absorption and  $2.0 \cdot 10^{-13}$  on fission — double precision round-off only.

**Throughput on RTX 3080.** On a  $10^6$ -energy log-spaced grid inside U-238’s RRR:

| Implementation                    | ns/lookup | Relative |
|-----------------------------------|-----------|----------|
| CPU single-thread (Ryzen 9800X3D) | 66.8      | 1.00×    |
| GPU (RTX 3080, saturated)         | 9.5       | 0.14×    |

7.0× per-lookup GPU-vs-CPU.

**Standalone GPU evaluator only.** The validation above is for the GPU evaluator on its own, exercised by a host driver that broadcasts a fixed energy grid to all threads and reads back the cross-section vector. Wiring the evaluator into the event-based `transport_persistent` kernel for end-to-end GPU hybrid transport requires a bitmap `has_wmp[i]` plus per-nuclide  $E_{\min}^{\text{WMP}}, E_{\max}^{\text{WMP}}$  checks inline, the recursive Humlicek conjugate fold unrolled to a flag-plus-sign-flip form (recursive `__device__` calls spill to local memory under the register pressure of `__launch_bounds__(256, 2)` and trigger `CUDA_ERROR_ILLEGAL_ADDRESS` on deep stacks), and a recomputed total cross section from partials with URR sampling suppressed inside the WMP window. We have not completed the in-kernel integration and report no in-engine GPU hybrid  $k_\infty$  or throughput.

## 11 Threats to validity

**Scope and nature of benchmarks.** Two systems: Godiva (HEU-MET-FAST-001, three nuclides, fast spectrum, *physical* ICSBEP benchmark with measured  $k_{\text{eff}}^{\text{exp}} = 1.0000 \pm 100$  pcm  $1\sigma$ ) and a nine-nuclide thermal PWR pin cell (no physical benchmark; we use OpenMC 0.15.3 as an independent code-to-code reference). Full-core LWR analyses carry  $\gtrsim 200$  nuclides; the hybrid memory picture scales with the reaction-channel mix and could look very different there. On Godiva we report  $\Delta_{\text{exp}}$  against ICSBEP as the primary validation metric and  $\Delta_{\text{OMC}}$  against OpenMC only as an independent cross-code check; the two codes straddle experiment from



opposite sides at this statistics, and the cross-code spread (178 pcm) is larger than either code’s offset from the measurement. On PWR, with no physical benchmark available, the OpenMC  $|\Delta| \leq 51$  pcm agreement is the strongest claim we can make and should be read as a code-to-code cross-check, not as absolute validation.

**S( $\alpha, \beta$ ) coverage.** Only H in H<sub>2</sub>O is exercised. D<sub>2</sub>O, graphite, ZrH, and other thermal libraries are not validated.

**Sampling convention.** Our original correlated-CDF sampling is a valid choice; we switched to OpenMC’s stochastic-bin convention because that is what we were comparing against. A fully independent reference (MCNP, Serpent) would reveal whether OpenMC itself is biased, which we do not investigate here.

**Charged-particle absorption.** Channels MT= 103, 107, first-chance fission (MT= 19–21) are absorbed into the capture macroscopic.  $k_\infty$  is correct under this convention; MT-level reaction-rate tallies are not.

**Statistical confidence.** All comparisons use  $\text{SEM} = \sigma_{\text{seed}} / \sqrt{N_{\text{seeds}}}$ , not  $\sigma_{\text{seed}}$  itself. Large-L3 sweeps use  $N_{\text{seeds}} = 10$ ; hybrid sweep uses five.

**Hybrid memory.** The 132.9× representation-byte ratio is not an in-engine reduction. In-engine reduction is 1.06× (Sec. 9.4). The 132.9× density ratio is reported for what it is; the 1.06× engine-level reduction is the number an engine builder would care about.

**GPU SVD.** Slower than GPU pointwise on PWR pin cell because of the Zr clad level-sum cost. A per-warp level-sum cache would likely fix this but is not implemented.

**GPU hybrid.** Only the WMP evaluator itself is validated on GPU; the transport-loop integration is not done. No in-engine GPU hybrid  $k_\infty$  or throughput is reported.

## 12 Related work

Adaptive resampling machine-learned models [4, 5] retain 10–50% of the pointwise data with  $\leq 97$  pcm criticality error. Our approach is orthogonal (replace the representation, not subsample the array). Windowed Multipole for Doppler broadening is due to [6, 7]; hybrid multipole + windowed networks in RMC code [8] pair multipole with a second analytical method; our hybrid pairs multipole with SVD. Event-based GPU transport follows the Tramm architecture [9] and shares the memory-bottleneck diagnosis of earlier XSBench studies [10]. Cross-isotope basis sharing was investigated and found not viable beyond the first singular vector (subspace angle  $> 85^\circ$  at  $k = 2$  between  $^{235}\text{U}$ ,  $^{238}\text{U}$ ,  $^{239}\text{Pu}$ ), a finding that echoes KV-cache compression experience in a different domain [13].

## 13 Conclusion

Truncated SVD of the  $\log_{10}$ -transformed cross-section matrix is a defensible drop-in *representation* for the pointwise table. On the Godiva HEU-MET-FAST-001 physical benchmark, our rank-5 SVD provider reproduces the measured critical  $k_{\text{eff}} = 1.0000 \pm 100$  pcm ( $1\sigma$ ) [2] to  $+79 \pm 38$  pcm — inside the experimental uncertainty band. On a nine-nuclide thermal PWR pin cell (no physical benchmark exists; OpenMC 0.15.3 is the independent code-to-code reference) the same engine running rank-5 SVD vs its own pointwise table agrees on  $k_\infty$  within

13–47 pcm on two hardware tiers, and all four in-engine providers (CPU table, CPU SVD, GPU pointwise, GPU SVD) agree with OpenMC 0.15.3 to  $|\Delta| \leq 51$  pcm at ten-seed statistics after a transport-level bug audit closed an initial 78–127 pcm engine offset. A Humlicek-W4 Faddeeva evaluator, pure-Rust WMP reader, and SVD+WMP hybrid provider all work end-to-end and agree with the ACE+WMP industry baseline within  $-26$  pcm at single-seed statistics ( $\ll 1\sigma_{\text{seed}}$ ); a CUDA port of the WMP evaluator is bit-exact against the CPU path and  $7\times$  faster per lookup on a consumer GPU.

On Godiva (HEU-MET-FAST-001) the physical benchmark is the ICSBEP experimental critical mass  $1.0000 \pm 100$  pcm ( $1\sigma$ ) [2]. Our rank-5 SVD provider reproduces it to  $+79 \pm 38$  pcm, inside the experimental uncertainty band. OpenMC 0.15.3 on the same ENDF/B-VII.1 HDF5 files sits at  $-99$  pcm, also inside the band; the two codes straddle experiment from opposite sides. Reaching that number required three transport sampling fixes —  $(n,2n)/(n,3n)$  secondary banking,  $\mu$ -CDF inversion for linear-linear ENDF angular distributions, and evaporation-vs-tabulated continuum-inelastic MT= 91 outgoing energy — which together close  $+325 \rightarrow +79$  pcm against ICSBEP (Sec. 5.1). Because OpenMC on the identical HDF5 sits inside  $\sigma_{\text{exp}}$  from the opposite side, the offset is engine-level, not a library bias. The cross-code spread to OpenMC ( $+178$  pcm at these statistics) is larger than either code’s offset from the experiment; we report it as a cross-code diagnostic, not as a deviation from the physical benchmark.

At a realistic *off-library* operating temperature, the story is more interesting. With a uniform  $+150$  K offset applied to every nuclide (fuel  $900 \rightarrow 1050$  K, moderator  $600 \rightarrow 750$  K), OpenMC-style stochastic  $T$ -picking enabled on the pointwise and hybrid paths, and a 3-point unity Ducru weight scheme on the SVD path (Sec. 2.4), SVD reproduces  $k_{\infty}$  within 213 pcm of ACE+WMP at ten-seed 50 000-particle 120-batch statistics — a small ( $\sim 3.6\sigma$ ) but real systematic, inside the publishable band. SVD is 22% smaller than WMP in memory and matches its speed within 0.07%. This is the result the paper exists to report: at a realistic operating temperature a compressed SVD representation of ENDF/B-VII.1 cross sections beats the industry ACE+WMP baseline on memory, ties it on speed, and trails it by a characterisable small systematic on  $k_{\infty}$  (Sec. 7.3).

Getting there was non-trivial. A naive 2-point unity Ducru leaves a 2094 pcm kernel-approximation bias at the same operating conditions (Sec. 7.2); a simplex-projected QP that clips the signed Ducru weights to non-negativity is *worse* (5% L2 error on U-238 capture versus 0.84% for the signed 3-point scheme) because the simplex projection is Euclidean-optimal in weight space whereas the raw Ducru weights are Euclidean-optimal in the Doppler-kernel norm (Sec. 7.5). We also found and fixed a channel-consistency bug in our stochastic  $T$ -pick: drawing a fresh  $\xi$  per partial channel produced non-monotonic  $k_{\infty}$ -versus- $T$  behaviour that on-library measurements cannot detect (Sec. 7.1). The PHYSOR-style off-library test surfaced it as a byproduct of stressing the representation.

The *performance* story on-library is less one-sided. CPU SVD rank-5 is  $1.37\times$ – $1.90\times$  faster than the in-engine pointwise table on PWR pin cell (hardware-dependent) and  $1.43\times$  faster than the table on Godiva on the large-L3 system; GPU SVD is  $1.30\times$  *slower* than GPU pointwise on PWR because four clad nuclides require summing 13 discrete-level SVD kernels per lookup. The hybrid is  $2.06\times$  slower than CPU SVD because WMP evaluation is  $\sim 15\times$  more expensive than the SVD FMA per lookup.

The *memory* story has two parts that must not be conflated. The representation-byte ratio of  $132.9\times$  (WMP + smooth-SVD vs four-channel six-temperature pointwise) is real; the live in-engine reduction from the smooth-only SVD rebuild is  $1.06\times$  ( $519.0 \rightarrow 487.6$  MB), and the hybrid engine is absolutely  $4.8$ – $5.2\times$  larger than the pointwise-table baseline. The discrete-level SVD kernels, which WMP does not cover, dominate the engine’s working set. Reducing those kernels from rank 5 to rank 1 — the reviewer’s suggestion and a free change because levels are weakly  $T$ -dependent (Sec. 7.4) — closes  $\approx 70\%$  of the engine’s SVD working set on Godiva with no measurable  $k_{\text{eff}}$  shift. This is what enables the off-library 22% memory win against

ACE+WMP referenced above: with discrete-level rank 1 applied, the SVD engine is absolutely smaller than the two-endpoint stochastic- $T$  pointwise table even before the WMP comparison.

The practical takeaway for a reader considering this approach: rank-5 SVD with 3-point unity Ducru temperature interpolation and discrete-level rank-1 is correct, reproduces the ICSBEP Godiva experimental  $k_{\text{eff}}$  within  $+79 \pm 38$  pcm (inside the 100 pcm experimental uncertainty), agrees with OpenMC and its own pointwise baseline within  $\leq 3 \text{ SEM}_{10}$  on PWR at on-library  $T$ , and agrees with ACE+WMP within  $\sim 3.6 \text{ SEM}_{10}$  on PWR at off-library  $T$ . On memory it wins modestly on-library and by 22% off-library; on speed it is  $1.4\times\text{--}1.9\times$  faster than pointwise on CPU and at parity with ACE+WMP off-library. The 43/47 rank-one reactions and the weakly- $T$ -dependent discrete levels are real structural properties of ENDF/B-VII.1 cross sections, worth exploiting; the raw Ducru weights are *not* a partition of unity and must be normalized before being applied to a  $\log \sigma$  SVD.

We implemented and measured the kernel-Gram QP from [1] §4 that we initially flagged as future work (Sec. 7.5) and found a structural subtlety worth reporting: at low rank it beats the signed unity scheme (U-238 MT= 102 6.67 eV peak ratio 1.000 at rank 3 vs 0.994 for signed unity), but at our production rank 5 it is *worse* (0.983 vs 0.990 on the same peak). The QP’s constrained optimum lies on the boundary of the probability simplex — one weight clips to zero, collapsing the 3-point problem back to a 2-point one — and at higher rank the signed unity scheme’s negative weight on the far-away library column is actively useful as a shape-cancellation term. Non-negativity is the wrong constraint once basis-truncation error stops dominating. This is the single highest-leverage piece of follow-up work we did not publish in the original version of this paper; the result is that signed 3-point unity *is* the right production default at rank 5, not merely a convenient approximation to the QP ideal. We report the positive, mixed, and negative findings together so the trade-off is visible.

## Data availability

The transport engine (pure Rust), the CUDA event-based solver, the CUDA WMP evaluator, all input decks, per-seed benchmark CSV files, the XS-audit scripts, the hybrid experiment Python script, and the scripts that generate the figures in this paper are released with the preprint. ENDF/B-VII.1 neutron and thermal data are available from the OpenMC data archive; the ENDF/B-VII.1 Windowed Multipole library is the public MIT release.

## Declaration of AI assistance

This work was written with assistance from large language models for code and prose drafting; all physics, numerical experiments, and results were reviewed and validated by the human author.

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